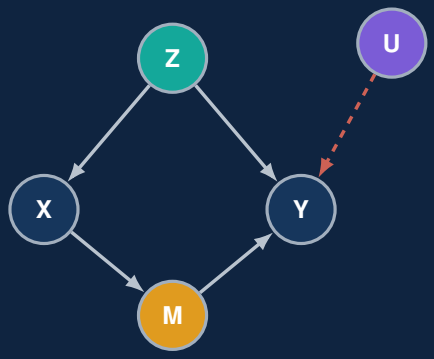


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 10

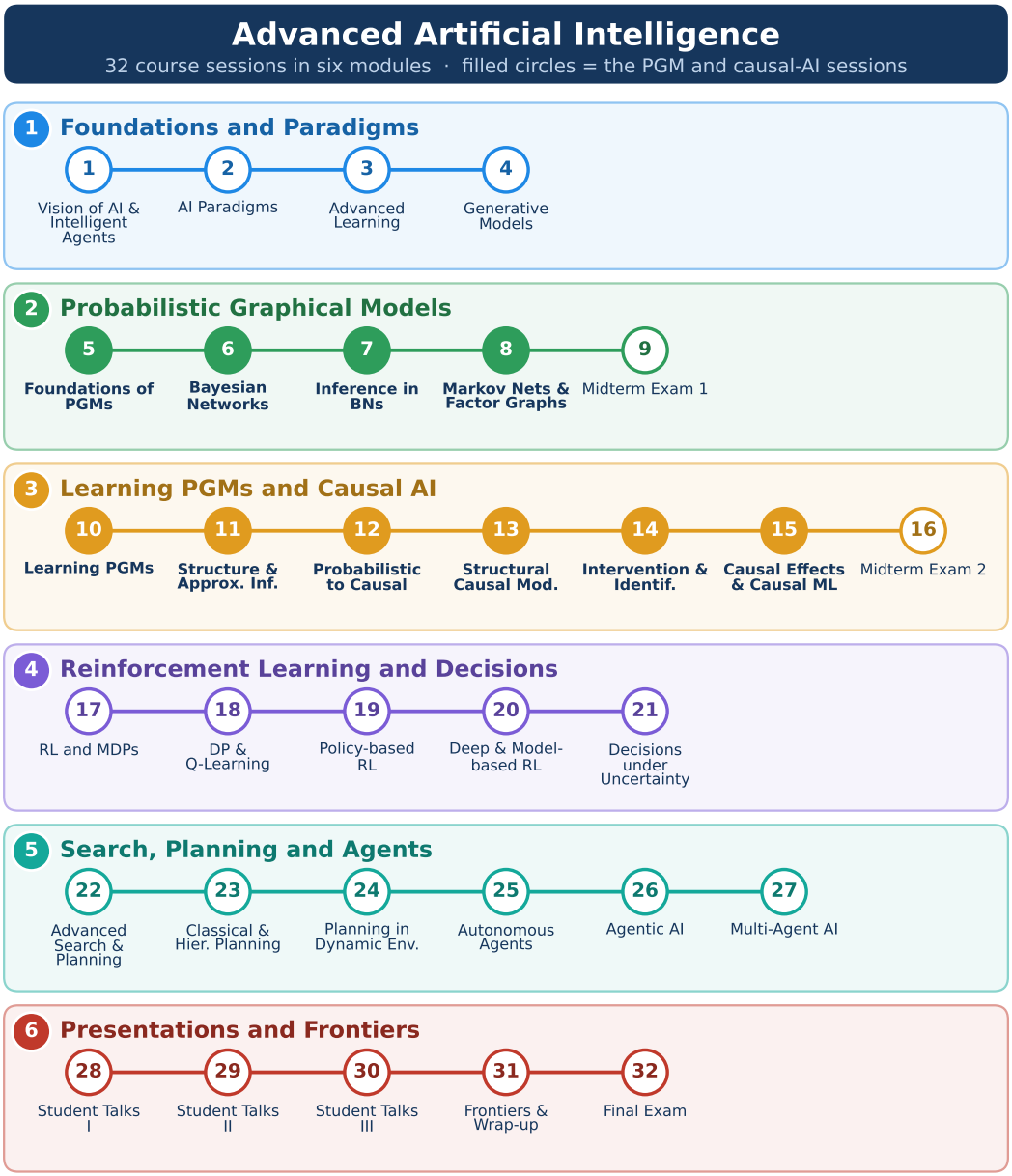
Learning: Parameter Estimation and Latent Variables

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 10 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, a "you are here" course map, the core theory organized in parts, a dedicated part on mathematical statistics, fully worked numerical and analytical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s10_examples.py` for the numbers, `python/s10_figures.py` for the figures), and `questions/s10_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions.

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SESSION 10 ADVANCED ARTIFICIAL INTELLIGENCE

Learning: Parameter Estimation and Latent Variables

Parameter learning, maximum likelihood, Bayesian learning and latent variables

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I
Maximum Likelihood
Decomposable likelihood, counts as sufficient statistics, cross-entropy.

PART II
Bayesian Learning
Beta and Dirichlet priors, smoothing, parameter independence.

PART III
Latent Variables & EM
Missing data, ELBO, EM for mixtures and for BNs.

PART IV
Mathematical Statistics
Fisher information, Cramér–Rao, intervals, MSE, EM rate.

PART V
AI Connections
LLMs, VAEs, model-based RL, bandits, agents, games.

LEARNING OBJECTIVES

- Derive the MLE of every CPT from the decomposable log-likelihood; identify family counts as sufficient statistics.
- Learn with Beta/Dirichlet priors: posterior, predictive, MAP, smoothing, parameter independence, online updates.
- Classify missing data as MCAR, MAR or MNAR and decide when the mechanism can be ignored.
- State, prove and run EM: the ELBO, Gaussian mixtures, expected counts in Bayesian networks.
- Quantify error: Fisher information, Cramér–Rao, asymptotic normality, Wilson intervals, shrinkage MSE, EM rate.
- Recognise these estimators in LLMs, VAEs, model-based RL, Thompson sampling, tool-using agents and games.

Where we are. Sessions 5–8 (course sessions 5–8) built the *representation* and *reasoning* (exact inference) halves of PGMs; course session 9 was the first midterm. Today we add the third pillar, *learning*: where do the numbers in the CPTs come from? Session 11 then learns the *graph itself* and replaces exact inference by sampling, which is also what the E-step of today's EM algorithm needs when the network is large.

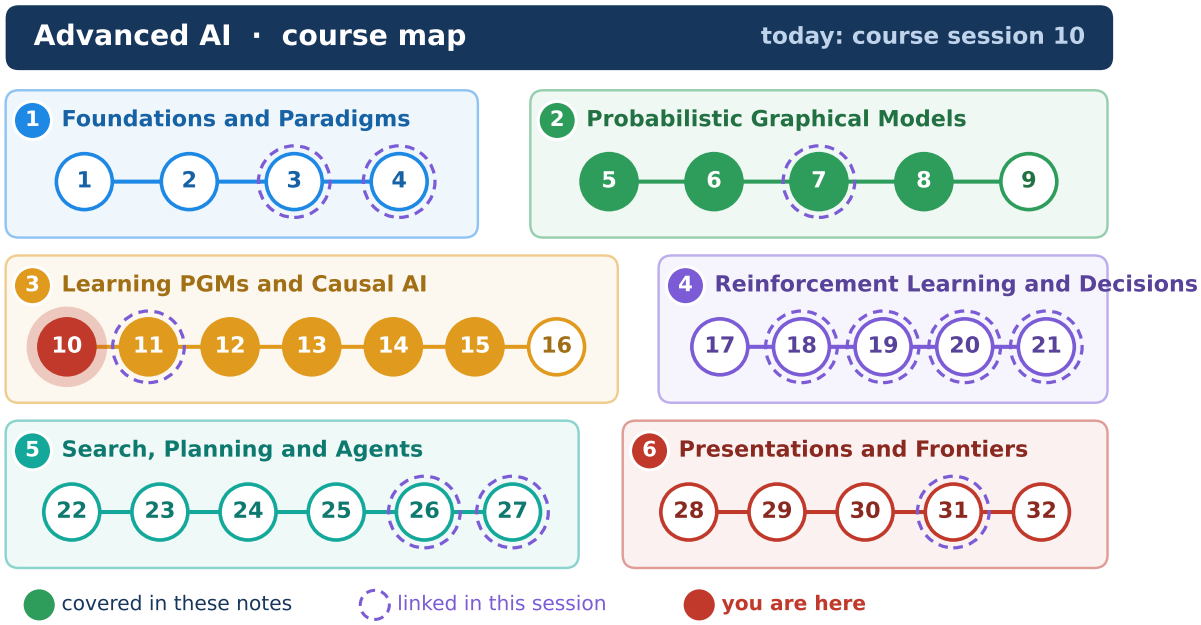


Figure 10.1. Where this session sits in the Advanced AI course (course session 10). Dashed purple rings mark the course sessions we link to today.

OPENING MOTIVATION

Every model in Sessions 6–8 assumed the numbers were simply handed to us: the student network's $P(\text{Grade} \mid \text{Difficulty}, \text{Intelligence})$, clique potentials, and so on. In reality, these numbers come from data. If we are lucky, every variable in our graph is observed in every data record, and estimating parameters is almost mechanical — a direct payoff of the factorization we spent two sessions establishing. But often the most useful variables in a model, like a customer's underlying "segment" or a patient's true (versus tested) disease status, are never directly observed. Today we handle both cases: maximum likelihood estimation when everything is observed, and the EM algorithm — built directly on Session 7's inference machinery — when it is not.

Healthcare

A diagnostic network learns $P(\text{test} \mid \text{disease})$ from hospital records in which the true disease status is often missing: EM with expected counts.

Robotics

A mobile robot learns its motion model $P(s' \mid s, a)$ by counting what happened after each command, with a Dirichlet prior so that unseen outcomes keep non-zero probability.

LLM agents

An assistant keeps a Beta posterior over how often each tool succeeds and routes calls by Thompson sampling; the LLM itself was trained by maximum likelihood.

PART I

Maximum Likelihood with Complete Data

Learning a Bayesian network is counting

PART I · MAXIMUM LIKELIHOOD

10.1 The Learning Problem

Notation $\mathcal{D} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}\}$: data $\quad \mathrm{pa}_i = \mathrm{pa}(X_i)$ $\quad u$: a value of pa_i $\quad \theta_{x|u} = P(X_i{=}x \mid \mathrm{pa}_i{=}u)$ $\quad N[x,u]$, $N[u]$: counts

Suppose we are given the DAG G of a Bayesian network over $X_1, \dots, X_n$ and a dataset $\mathcal{D} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}\}$ of N fully-observed i.i.d. samples. We want $\hat{\theta} = \arg\max_\theta \; P(\mathcal{D} \mid \theta)$. The *likelihood* $L(\theta; \mathcal{D}) = \prod_j P(\mathbf{x}^{(j)}; \theta)$ is the probability of the observed data viewed as a function of the parameters, and $\ell = \log L$ is the log-likelihood. The i.i.d. assumption is itself a conditional-independence statement: the records are independent *given* θ (Session 5).

PART I · MAXIMUM LIKELIHOOD

10.2 The Decomposable Likelihood and Sufficient Statistics

Proposition 10.1 (Decomposable log-likelihood)**.** *For a Bayesian network with parameters $\theta = (\theta_1, \dots, \theta_n)$, where θ_i parameterizes $P(x_i \mid \mathrm{pa}(x_i); \theta_i)$, the log-likelihood decomposes as*

$$\ell(\theta; \mathcal{D}) = \log \prod_{j=1}^{N} \prod_{i=1}^{n} P\big(x_i^{(j)} \mid \mathrm{pa}(x_i)^{(j)}; \theta_i\big) = \sum_{i=1}^{n} \underbrace{\sum_{j=1}^{N} \log P\big(x_i^{(j)} \mid \mathrm{pa}(x_i)^{(j)}; \theta_i\big)}_{\ell_i(\theta_i; \mathcal{D})} .$$

This is an immediate consequence of the DAG factorization from Session 6, combined with log turning products into sums. Its significance is entirely practical: if the parameter sets $\theta_1, \dots, \theta_n$ do not overlap (the usual case), *maximizing $\ell(\theta; \mathcal{D})$ decomposes into n separate, independent optimization problems*, one per node, each involving only that node and its parents.

Consequence: closed-form CPT estimates. For discrete CPTs, each $\ell_i(\theta_i)$ is maximized, subject to each row of the CPT summing to one, at the empirical conditional frequency

$$\hat{P}(X_i = v \mid \mathrm{pa}_i = u) = \frac{\mathrm{count}(X_i{=}v,\ \mathrm{pa}_i{=}u)}{\mathrm{count}(\mathrm{pa}_i{=}u)} = \frac{N[v,u]}{N[u]}.$$

The proof is Theorem 10.1 below (one line via Gibbs' inequality, plus the Lagrange-multiplier derivation).

INTUITION

Learning a Bayesian network with full observability reduces to counting: for every node, tally how often each value occurred alongside each parent configuration, and normalize. This is only possible *because* the graph told us, via the local Markov property, exactly which variables are relevant to each CPT — a smaller graph means fewer counts to accumulate and less data needed per parameter, directly mirroring Session 5's compactness argument.

For discrete CPTs, grouping the records by family value gives the *multinomial form* of the likelihood

$$L(\theta; \mathcal{D}) = \prod_{i=1}^{n} \prod_{u \in \mathrm{Val}(\mathrm{pa}_i)} \prod_{x \in \mathrm{Val}(X_i)} \theta_{x|u}^{N[x,u]}, \qquad N[x,u] = \sum_{j=1}^{N} \mathbb{1}\{x_i^{(j)} = x,\ \mathrm{pa}_i^{(j)} = u\}. \tag{10.1}$$

Each CPT *row* $\theta_{\cdot|u}$ appears in its own factor: the likelihood decomposes not only over nodes but also over parent configurations ("local decomposability"). For binary $A \to B$ with $\theta_0 = P(B{=}1 \mid A{=}0)$ and $\theta_1 = P(B{=}1 \mid A{=}1)$, the log-likelihood of $P(B \mid A)$ is $\ell_0(\theta_0) + \ell_1(\theta_1)$ with $\ell_a(\theta_a) = N[1,a]\log\theta_a + N[0,a]\log(1-\theta_a)$: a sum of two one-dimensional functions with no cross term (its Hessian is diagonal), so each row is maximized on its own. In short, all three learning regimes of this session (complete data, priors, hidden data) reduce to *(expected) family counts.*

Proposition 10.2 (Family counts are sufficient)**.** *The collection of family counts* $T(\mathcal{D}) = \{N[x,u]\}_{i,x,u}$ *is a sufficient statistic for* θ*; in fact it is minimal sufficient when all CPT rows are free.*

Proof. By (10.1), $P(\mathcal{D};\theta) = g(T(\mathcal{D}),\theta)\,h(\mathcal{D})$ with $h \equiv 1$, so the Fisher–Neyman factorization theorem gives sufficiency. Minimality: for two datasets with counts N and N', $P(\mathcal{D};\theta)/P(\mathcal{D}';\theta) = \prod_{i,u}\prod_x \theta_{x|u}^{d_{x,u}}$ with $d_{x,u} = N[x,u] - N'[x,u]$. The rows are free and vary independently, so the ratio is free of θ only if every row factor $\prod_x \theta_{x|u}^{d_{x,u}}$ is constant on the open simplex. Taking logs, $\sum_x d_{x,u}\log\theta_{x|u}$ constant there forces all $d_{x,u} = 0$ (e.g. for a binary row, $d_1\log\theta + d_0\log(1-\theta)$ is constant in $\theta \in (0,1)$ only if $d_1 = d_0 = 0$). Hence the ratio is free of θ iff $N = N'$. □

Theorem 10.1 (MLE of a CPT)**.** *If* $N[u] = \sum_x N[x,u] > 0$*, the unique maximizer of* ℓ *over the probability simplex of row* u *is* $\hat\theta_{x|u} = N[x,u]/N[u]$*.*

Proof. The row contributes $\ell_u(\theta) = \sum_x N[x,u]\log\theta_{x|u} = N[u]\sum_x \hat p_x \log\theta_{x|u}$ with $\hat p_x = N[x,u]/N[u]$. By Gibbs' inequality, $\sum_x \hat p_x\log\theta_x \le \sum_x \hat p_x\log\hat p_x$ with equality iff $\theta = \hat p$, because the difference is $-\mathrm{KL}(\hat p\,\|\,\theta) \le 0$.

Same result by a Lagrange multiplier. Maximize $\sum_x N[x,u]\log\theta_x + \lambda\big(1 - \sum_x \theta_x\big)$. Setting the derivative in θ_x to zero gives $N[x,u]/\theta_x = \lambda$, i.e. $\theta_x = N[x,u]/\lambda$; summing over x and using $\sum_x \theta_x = 1$ gives $\lambda = \sum_x N[x,u] = N[u]$, hence $\theta_x = N[x,u]/N[u]$. □

Worked Example 10.1 (Counting CPTs for the student network)**.** Suppose 200 fully-observed student records give: among the 80 students with difficulty D = hard, 32 have intelligence I = high; among the 120 with D = easy, 60 have I = high. In the student network of Session 6, D and I are root nodes (no parents, hence marginally independent), so their CPTs are the marginals $P(D)$ and $P(I)$, not $P(I \mid D)$. Counting gives $\hat P(D{=}\text{hard}) = 80/200 = 0.40$ and $\hat P(I{=}\text{high}) = (32+60)/200 = 92/200 = 0.46$. For the grade CPT $P(\text{Grade} \mid D, I)$ we would similarly tally grade outcomes within each of the four (D,I) combinations and normalize each of those four rows separately — exactly Proposition 10.1's claim that each CPT is estimated independently of every other one.

PYTHON LAB · `s10_examples.py` · `ex_counting_student()`

Prints `P_hat(I=high) = (32+60)/200 = 0.46` and `P_hat(D=hard) = 80/200 = 0.40`.

Remark 10.1 (Parameter sharing). In an HMM the transition CPT is shared by all time steps; counts are simply *pooled*, $N[s',s] = \sum_t \mathbb{1}\{s_{t+1} = s', s_t = s\}$. When the shared CPD is a neural network (an LLM, Part V) the closed form is lost and we use gradient ascent.

Worked Example 10.2 (MLE for a linear-Gaussian CPD (analytical))**.** Let $X \mid \mathrm{pa}(X){=}\mathbf{u} \sim N(\beta_0 + \boldsymbol{\beta}^\top \mathbf{u}, \sigma^2)$ (Session 6's linear-Gaussian BN). Collect the coefficients in $\beta = (\beta_0, \boldsymbol{\beta}^\top)^\top \in \mathbb{R}^p$, where $p = 1 + |\mathrm{pa}(X)|$, and let U be the $N \times p$ design matrix with rows $(1, \mathbf{u}_j^\top)$ (assumed of full column rank) and $\mathbf{x} = (x_1, \ldots, x_N)^\top$ the responses. Then $\ell_X(\beta, \sigma^2) = -\frac{N}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\mathbf{x} - U\beta\|^2$.

1. For fixed σ^2, only the last term depends on β: $\nabla_\beta \ell_X = \frac{1}{\sigma^2} U^\top(\mathbf{x} - U\beta) = 0$ gives the normal equations $U^\top U\beta = U^\top \mathbf{x}$, so $\hat\beta = (U^\top U)^{-1} U^\top \mathbf{x}$, the ordinary least-squares estimator.
2. With $\mathrm{RSS} = \|\mathbf{x} - U\hat\beta\|^2$ (residual sum of squares), $\partial \ell_X / \partial \sigma^2 = -\frac{N}{2\sigma^2} + \frac{\mathrm{RSS}}{2\sigma^4} = 0$ gives $\hat\sigma^2 = \mathrm{RSS}/N$.

Since $\|\mathbf{x} - U\beta\|^2 = \mathbf{x}^\top\mathbf{x} - 2\beta^\top U^\top \mathbf{x} + \beta^\top U^\top U\beta$, the sufficient statistics are $U^\top U$, $U^\top \mathbf{x}$ and $\mathbf{x}^\top \mathbf{x}$: the continuous analogue of family counts.

PART I · MAXIMUM LIKELIHOOD

10.3 Maximum Likelihood as Cross-Entropy Minimization

Proposition 10.3 (MLE minimizes KL to the empirical distribution)**.** *Let $\hat p_{\mathcal{D}}$ be the empirical distribution of the data. Then $\frac{1}{N}\ell(\theta; \mathcal{D}) = \sum_{\mathbf{x}} \hat p_{\mathcal{D}}(\mathbf{x}) \log p_\theta(\mathbf{x}) = -H(\hat p_{\mathcal{D}}) - \mathrm{KL}(\hat p_{\mathcal{D}} \,\|\, p_\theta)$, so $\arg\max_\theta \ell = \arg\min_\theta \mathrm{KL}(\hat p_{\mathcal{D}} \| p_\theta) = \arg\min_\theta H(\hat p_{\mathcal{D}}, p_\theta)$ (cross-entropy).*

Proof. Group equal records: $\frac{1}{N}\sum_j \log p_\theta(\mathbf{x}^{(j)}) = \sum_{\mathbf{x}} \hat p(\mathbf{x}) \log p_\theta(\mathbf{x})$, then add and subtract $\sum \hat p \log \hat p$. □

Worked Example 10.3 (Cross-entropy of three candidate models)**.** A three-valued variable is observed $N = 10$ times with counts $(5, 3, 2)$, so $\hat p_{\mathcal{D}} = (0.5, 0.3, 0.2)$. Cross-entropy $H(\hat p, q) = -\sum_k \hat p_k \log q_k$ (natural logs, in nats) for three models q, term by term:

model q	terms $-\hat p_k \log q_k$, $k = 1, 2, 3$	$H(\hat p, q)$	$\mathrm{KL}(\hat p \,\|\, q)$
$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$	$0.5493 + 0.3296 + 0.2197$	1.0986	0.0690
$(0.4, 0.4, 0.2)$	$0.4581 + 0.2749 + 0.3219$	1.0549	0.0253
$\hat p = (0.5, 0.3, 0.2)$	$0.3466 + 0.3612 + 0.3219$	1.0297	0

(e.g. $-0.5 \log 0.4 = 0.4581$.) The minimum, $1.0297 = H(\hat p)$, is attained at the MLE, and the excess over $H(\hat p)$ is the KL divergence: $1.0986 - 1.0297 = 0.0690$ and $1.0549 - 1.0297 = 0.0253$ (up to rounding in the last digit), exactly as Proposition 10.3 states. The average log-likelihood per record is $-H$.

PYTHON LAB · `s10_examples.py` · `ex_kl_cross_entropy()`

```
cross-entropy = 1.0986 KL = 0.0690;  1.0549 KL = 0.0253;  1.0297 KL = 0.0000;
entropy of p_emp = 1.0297.
```

COMMON PITFALL A zero count gives $\hat{\theta}_{x|u} = 0$: any future record containing (x, u) gets log-likelihood $-\infty$. With many parent configurations zero counts are the rule. Part II fixes this with priors.

PART II

Bayesian Parameter Learning

Parameters are random variables too

10.4 The Beta–Bernoulli Model

In the Bayesian view θ is itself a random variable with a prior $p(\theta)$, and learning is inference: $p(\theta \mid \mathcal{D}) \propto p(\theta)L(\theta; \mathcal{D})$. Graphically, θ becomes a parent of every record (Fig. 10.3), and the records are conditionally independent given θ but *not* marginally independent: each record carries information about the next.

Theorem 10.2 (Beta–Bernoulli conjugacy). *Let* $\theta \sim \mathrm{Beta}(\alpha, \beta)$ *and* $X_1, \dots, X_n \mid \theta \overset{iid}{\sim} \mathrm{Bernoulli}(\theta)$ *with* $k = \sum_i X_i$. *Then*

(a) $\theta \mid \mathcal{D} \sim \mathrm{Beta}(\alpha + k,\ \beta + n - k)$;

(b) posterior predictive: $P(X_{n+1} = 1 \mid \mathcal{D}) = \mathbb{E}[\theta \mid \mathcal{D}] = \dfrac{\alpha + k}{\alpha + \beta + n}$;

(c) posterior mean $= w_n \dfrac{k}{n} + (1 - w_n)\dfrac{\alpha}{\alpha + \beta}$ *with* $w_n = \dfrac{n}{n + \alpha + \beta}$; *MAP* $= \dfrac{\alpha + k - 1}{\alpha + \beta + n - 2}$ *when* $\alpha + k > 1,\ \beta + n - k > 1$.

Proof. (a) $p(\theta \mid \mathcal{D}) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1} \cdot \theta^k(1-\theta)^{n-k}$, the kernel of $\mathrm{Beta}(\alpha + k, \beta + n - k)$. (b) $P(X_{n+1}{=}1 \mid \mathcal{D}) = \int \theta\, p(\theta \mid \mathcal{D})\, d\theta$ because $X_{n+1} \perp\!\!\!\perp \mathcal{D} \mid \theta$; the mean of $\mathrm{Beta}(a, b)$ is $a/(a+b)$. (c) Split the numerator of (b): $\frac{\alpha+k}{\alpha+\beta+n} = \frac{n}{\alpha+\beta+n} \cdot \frac{k}{n} + \frac{\alpha+\beta}{\alpha+\beta+n} \cdot \frac{\alpha}{\alpha+\beta} = w_n \frac{k}{n} + (1-w_n)\frac{\alpha}{\alpha+\beta}$. The MAP is the mode $(a{-}1)/(a{+}b{-}2)$ of $\mathrm{Beta}(a, b)$, obtained by setting the derivative of $(a{-}1)\log\theta + (b{-}1)\log(1{-}\theta)$ to zero. □

The prior acts as $\alpha + \beta$ *pseudo-observations*, the equivalent sample size: the posterior mean is a convex combination of the MLE and the prior mean, with weight $w_n \to 1$ as data accumulate.

Proposition 10.4 (Sequential = batch updating). *If records are conditionally i.i.d. given* θ, *then for any split* $\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2$, $p(\theta \mid \mathcal{D}) \propto p(\theta \mid \mathcal{D}_1)\, P(\mathcal{D}_2 \mid \theta)$: *yesterday's posterior is today's prior.*

Proof. $p(\theta \mid \mathcal{D}) \propto p(\theta)P(\mathcal{D}_1 \mid \theta)P(\mathcal{D}_2 \mid \theta) \propto p(\theta \mid \mathcal{D}_1)P(\mathcal{D}_2 \mid \theta)$, using $\mathcal{D}_1 \perp\!\!\!\perp \mathcal{D}_2 \mid \theta$. □

For conjugate models this is "add the new counts to the hyperparameters": an *online* algorithm with $O(1)$ memory per CPT row.

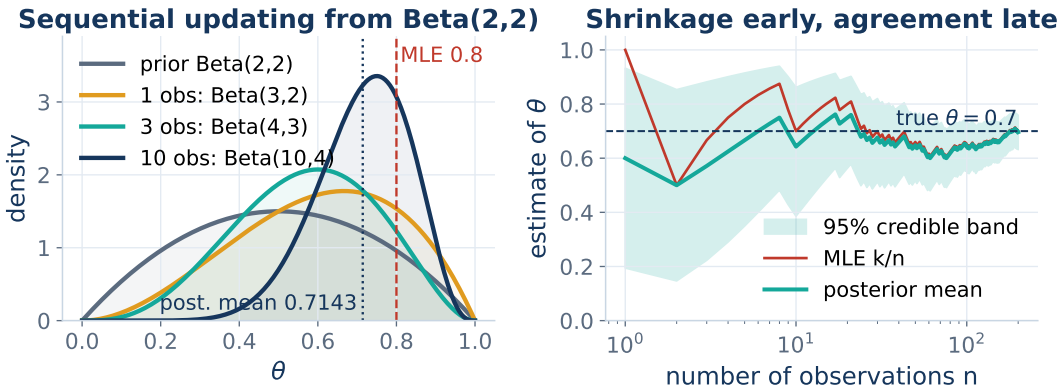


Figure 10.2. Left: Beta$(2,2)$ prior updated by the stream $1,1,0,1,1,1,0,1,1,1$ of Worked Example 10.4; after ten records the posterior is Beta$(10,4)$ whatever the order. Right: a simulated stream of 200 Bernoulli(0.7) records (seed 12 in `s10_figures.py`, `fig_beta_updates()`) with the same Beta$(2,2)$ prior: the posterior mean $(2+k)/(4+n)$ is pulled towards the prior mean $1/2$ early on and agrees with the MLE k/n later; the band is the 95% equal-tailed credible interval.

Worked Example 10.4 (Sequential Beta updating)**.** Prior Beta(α,β) = Beta$(2,2)$; data $1,1,0,1,1,1,0,1,1,1$, so $n=10$ and $k=8$.

1. *Sequential.* Each 1 adds one to the first parameter, each 0 to the second: after record 1 ($x=1$) Beta$(3,2)$; after records 2–3 $(1,0)$ Beta$(4,3)$; after all ten, Beta$(2+8,\,2+2)$ = Beta$(10,4)$, the same as the batch update of Theorem 10.2(a), as Proposition 10.4 guarantees.
2. *Point estimates.* Posterior mean $10/14 = 0.7143$; MAP $(10-1)/(10+4-2) = 9/12 = 0.75$; MLE $k/n = 8/10 = 0.8$.
3. *Shrinkage form.* $w_n = n/(n+\alpha+\beta) = 10/14 = 0.7143$ and the prior mean is $2/4 = 0.5$: $0.7143 \times 0.8 + 0.2857 \times 0.5 = 0.5714 + 0.1429 = 0.7143$.
4. *Credible interval.* The 2.5% and 97.5% quantiles of Beta$(10,4)$ (no closed form; computed with `scipy.stats.beta.ppf`) give the 95% equal-tailed interval $(0.4619,\ 0.9091)$.

PYTHON LAB · `s10_examples.py` · `ex_beta_sequence()`

`after 1 obs: Beta(3,2), after 3 obs: Beta(4,3), after 10 obs: Beta(10,4); batch update: Beta(10,4) (identical)`; mean 0.7143, MAP 0.7500, CI $(0.4619, 0.9091)$.

PART II · BAYESIAN LEARNING

10.5 Dirichlet Priors for CPTs

Definition 10.1 (Dirichlet distribution)**.** $\theta = (\theta_1,\dots,\theta_K)$ on the simplex has a Dir$(\alpha_1,\dots,\alpha_K)$ distribution, $\alpha_k > 0$, if $p(\theta) = \frac{\Gamma(\alpha_0)}{\prod_k \Gamma(\alpha_k)} \prod_k \theta_k^{\alpha_k - 1}$, $\alpha_0 = \sum_k \alpha_k$. Its mean is α_k/α_0 and for $K=2$ it is the Beta distribution.

Theorem 10.3 (Dirichlet–multinomial conjugacy)**.** *If $\theta \sim \mathrm{Dir}(\boldsymbol{\alpha})$ and the counts are $\mathbf{N} = (N_1,\dots,N_K)$, $N = \sum_k N_k$, then $\theta \mid \mathcal{D} \sim \mathrm{Dir}(\boldsymbol{\alpha}+\mathbf{N})$, the predictive probability of category k is*

$(\alpha_k + N_k)/(\alpha_0 + N)$, *and the MAP is* $(\alpha_k + N_k - 1)/(\alpha_0 + N - K)$ *whenever all* $\alpha_k + N_k > 1$.

Proof. Prior $\times$ likelihood $\propto \prod_k \theta_k^{\alpha_k - 1} \prod_k \theta_k^{N_k} = \prod_k \theta_k^{\alpha_k + N_k - 1}$. The predictive is the posterior mean. For the mode, maximize $\sum_k (a_k - 1) \log \theta_k$ on the simplex with a Lagrange multiplier as in Theorem 10.1, with $a_k - 1$ in place of $N[x, u]$. □

Laplace smoothing is a Dirichlet prior. With $\alpha_k = 1$ (uniform prior) the predictive is $(N_k + 1)/(N + K)$, Laplace's "add-one" rule; $\alpha_k = \frac{1}{2}$ is the Jeffreys prior ("add one half"). Note the subtle identity: the *MAP* under $\mathrm{Dir}(2, \dots, 2)$ equals the *posterior mean* under $\mathrm{Dir}(1, \dots, 1)$.

Worked Example 10.5 (Smoothing a CPT row with a zero count)**.** A diagnostic network records the symptom level (none, mild, severe) of 10 patients with a given disease: counts $(6, 4, 0)$. Here $N = 10$ and $K = 3$. The MLE $(6, 4, 0)/10 = (0.6, 0.4, 0)$ declares a severe symptom impossible. By Theorem 10.3:

1. $\mathrm{Dir}(1, 1, 1)$, $\alpha_0 = 3$: posterior mean $(6 + 1, 4 + 1, 0 + 1)/(10 + 3) = (7, 5, 1)/13 = (0.5385, 0.3846, 0.0769)$.
2. Jeffreys $\mathrm{Dir}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, $\alpha_0 = 1.5$: posterior mean $(6.5, 4.5, 0.5)/11.5 = (0.5652, 0.3913, 0.0435)$.
3. MAP under $\mathrm{Dir}(2, 2, 2)$, $\alpha_0 = 6$: $(6+2-1, 4+2-1, 0+2-1)/(10+6-3) = (7, 5, 1)/13$, again $(0.5385, 0.3846, 0.0769)$.

PYTHON LAB · `s10_examples.py` · `ex_dirichlet_smoothing()`

```
MLE [0.6 0.4 0.];  posterior mean Dir(1.0) [0.5385 0.3846 0.0769];  Dir(0.5)
[0.5652 0.3913 0.0435]; MAP under Dir(2,2,2) [0.5385 0.3846 0.0769].
```

10.5.1 Parameter independence and the Bayesian network as a whole

Definition 10.2 (Global and local parameter independence)**.** A prior over the parameters of a BN satisfies *global parameter independence* if $p(\theta) = \prod_i p(\theta_{X_i|\mathrm{pa}_i})$, and *local parameter independence* if moreover $p(\theta_{X_i|\mathrm{pa}_i}) = \prod_u p(\theta_{X_i|u})$ (CPT rows independent). This is the standard assumption of Heckerman, Geiger and Chickering (1995).

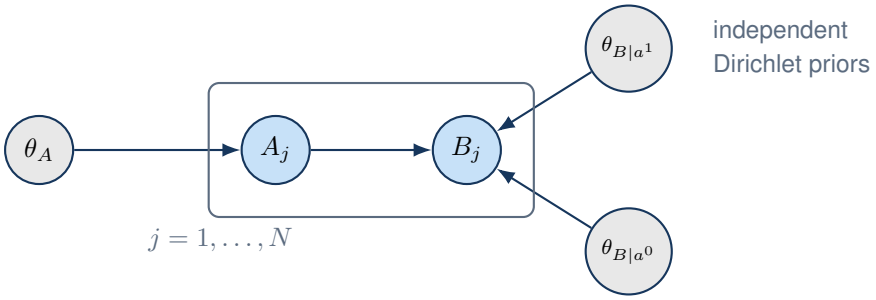


Figure 10.3. Meta-network of Bayesian learning for binary $A \to B$: the unknown parameters $\theta_A = P(A{=}1)$ and the two CPT rows $\theta_{B|a^1}, \theta_{B|a^0}$ are nodes (grey: unobserved), and the plate repeats the records (A_j, B_j) (blue: observed). With complete data the parameters stay independent a posteriori (Theorem 10.4); if A_j is missing, the observed collider B_j couples them.

Theorem 10.4 (Posterior parameter independence)**.** *Under global and local parameter independence with Dirichlet priors $\theta_{X_i|u} \sim \mathrm{Dir}(\boldsymbol{\alpha}_{i,u})$ and complete data, the posterior factorizes as $p(\theta \mid \mathcal{D}) = \prod_i \prod_u \mathrm{Dir}\big(\theta_{X_i|u};\ \boldsymbol{\alpha}_{i,u} + \mathbf{N}_{i,u}\big)$, where $\mathbf{N}_{i,u} = (N[x,u])_x$.*

Proof. By (10.1) the likelihood is a product of factors $\prod_x \theta_{x|u}^{N[x,u]}$, one for each pair (i,u), and by assumption so is the prior. The product of the two is therefore $\prod_{i,u} \big[p(\theta_{X_i|u}) \prod_x \theta_{x|u}^{N[x,u]}\big]$; each bracket depends on one block only, so the blocks are independent a posteriori, and each bracket is the Dirichlet kernel of Theorem 10.3. □

Proposition 10.5 (Bayesian prediction = plug in posterior means)**.** *Under the conditions of Theorem 10.4, $P(\mathbf{x}^{(N+1)} \mid \mathcal{D}) = \prod_i \dfrac{\alpha_{x_i|u_i} + N[x_i,u_i]}{\alpha_{u_i} + N[u_i]}$, where u_i is the parent configuration of X_i in the new record and $\alpha_u = \sum_x \alpha_{x|u}$.*

Proof. $P(\mathbf{x}^{(N+1)} \mid \mathcal{D}) = \int \prod_i \theta_{x_i|u_i}\, p(\theta \mid \mathcal{D})\, d\theta$. Each factor involves a different block (i,u_i) and the blocks are independent, so the integral is the product of the posterior means $\mathbb{E}[\theta_{x_i|u_i} \mid \mathcal{D}]$. □

Worked Example 10.6 (Bayesian prediction in $A \to B$)**.** Take the $N = 10$ complete records of Worked Example 10.13, $(A,B) = (1,1) \times 4,\ (1,0) \times 1,\ (0,1) \times 2,\ (0,0) \times 3$, and independent $\mathrm{Dir}(1,1)$ (uniform) priors on θ_A, $\theta_{B|a^1}$ and $\theta_{B|a^0}$. The counts are $N[A{=}1] = 5$, $N[A{=}0] = 5$ and $N[B{=}1, A{=}1] = 4$. By Proposition 10.5,

$$P(A{=}1 \mid \mathcal{D}) = \frac{1+5}{2+10} = \frac{6}{12} = 0.5000, \qquad P(B{=}1 \mid A{=}1, \mathcal{D}) = \frac{1+4}{2+5} = \frac{5}{7} = 0.7143,$$

so $P(A{=}1, B{=}1 \mid \mathcal{D}) = 0.5000 \times 0.7143 = 0.3571$, slightly below the plug-in MLE value $0.5 \times 0.8 = 0.4$.

PYTHON LAB · `s10_examples.py` · `ex_bayes_prediction_bn()`

```
P(A=1 | D) = 6/12 = 0.5000; P(B=1 | A=1, D) = 5/7 = 0.7143; P(A=1, B=1 | D) = 0.3571.
```

Table 10.1. Three point estimates of a CPT row from the same counts N_k (prior $\mathrm{Dir}(\boldsymbol{\alpha})$).

	MLE	MAP	Posterior mean
Formula	N_k/N	$(N_k+\alpha_k-1)/(N+\alpha_0-K)$	$(N_k+\alpha_k)/(N+\alpha_0)$
Zero counts	probability 0	0 if $\alpha_k \le 1$	always > 0
Optimal under	— (frequentist)	0–1 loss	squared loss; predictive
Large N	consistent	$\to$ MLE	$\to$ MLE, same CLT

PART III

Missing Data, Latent Variables and EM

When the log of a sum refuses to split

PART III · LATENT VARIABLES

10.6 Missing Data Mechanisms

Let $\mathbf{Y}$ be the full data, split into observed and missing parts $(\mathbf{Y}_{\text{obs}}, \mathbf{Y}_{\text{mis}})$, and let $\mathbf{R}$ be the vector of missingness indicators ($R = 1$ if the entry is missing). The data model is $p(\mathbf{y} \mid \theta)$ and the *missingness mechanism* is $p(\mathbf{r} \mid \mathbf{y}, \psi)$ (Rubin, 1976; Little and Rubin, 2002).

Definition 10.3 (MCAR, MAR, MNAR)**.** The data are *missing completely at random* (MCAR) if $p(\mathbf{r} \mid \mathbf{y}, \psi) = p(\mathbf{r} \mid \psi)$; *missing at random* (MAR) if $p(\mathbf{r} \mid \mathbf{y}, \psi) = p(\mathbf{r} \mid \mathbf{y}_{\text{obs}}, \psi)$; and *missing not at random* (MNAR) otherwise. In graphical terms: MCAR means R has no parents among the variables, MAR means $R \perp\!\!\!\perp \mathbf{Y}_{\text{mis}} \mid \mathbf{Y}_{\text{obs}}$, MNAR means R depends on the value that is missing.

Theorem 10.5 (Ignorability under MAR)**.** *If the data are MAR and θ and ψ are variation independent (distinct), then the likelihood of the observed data factorizes as $L(\theta, \psi; \mathbf{y}_{obs}, \mathbf{r}) = p(\mathbf{r} \mid \mathbf{y}_{obs}, \psi) \cdot p(\mathbf{y}_{obs} \mid \theta)$; hence the MLE of θ (and the posterior of θ under independent priors) can be computed from $p(\mathbf{y}_{obs} \mid \theta) = \int p(\mathbf{y}_{obs}, \mathbf{y}_{mis} \mid \theta)\, d\mathbf{y}_{mis}$ alone.*

Proof. $L = \int p(\mathbf{y}_{\text{obs}}, \mathbf{y}_{\text{mis}} \mid \theta)\, p(\mathbf{r} \mid \mathbf{y}_{\text{obs}}, \mathbf{y}_{\text{mis}}, \psi)\, d\mathbf{y}_{\text{mis}}$. Under MAR the second factor does not depend on $\mathbf{y}_{\text{mis}}$ and leaves the integral; the remaining integral is $p(\mathbf{y}_{\text{obs}} \mid \theta)$. The maximization over (θ, ψ) then separates because the parameter space is a product. □

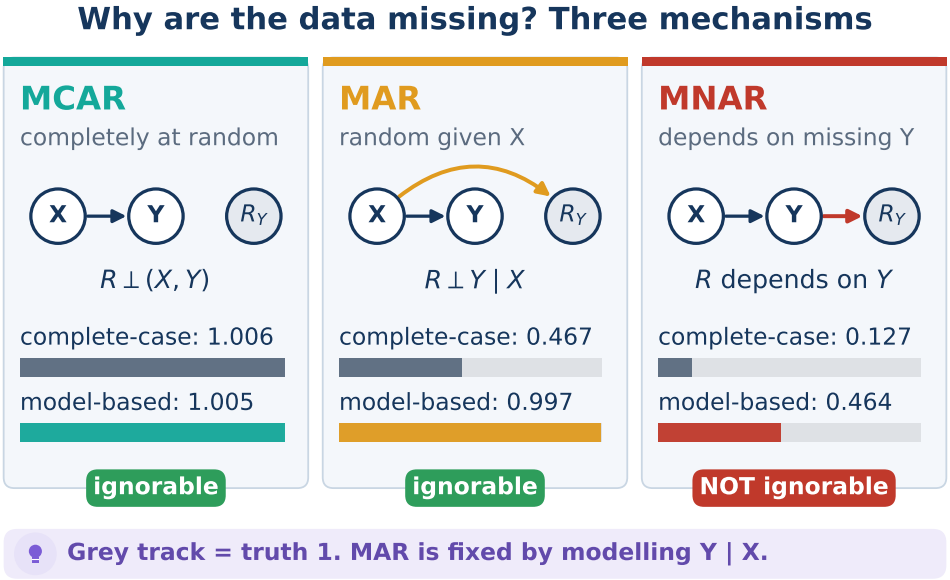


Figure 10.4. The three missing-data mechanisms as small DAGs with a missingness indicator R_Y, and the simulation of Worked Example 10.7 (true $\mathbb{E}[Y] = 1$): the complete-case mean of Y is fine under MCAR (1.006), biased under MAR (0.467) but repaired by modelling $Y \mid X$ with all X (0.997), and biased under MNAR either way (0.127, 0.464). Bars show each estimate as a fraction of a grey track of length 1.

Worked Example 10.7 (Complete cases versus a model for $Y \mid X$ (simulation))**.** *Setup* (`ex_missing_mechanisms()`, seed 5): $n = 200\,000$ records with $X \sim N(0, 1)$, $Y = 1 + X + \varepsilon$, $\varepsilon \sim N(0, 1)$, so the target is $\mathbb{E}[Y] = 1$. X is always observed; Y is deleted with probability 0.4

(MCAR), $\sigma(1.5X)$ (MAR) or $\sigma(1.5(Y-1))$ (MNAR), where $\sigma(z) = 1/(1+e^{-z})$; this removes 39.9%, 50.0% and 49.9% of the Y values. *Two estimators.* (i) Complete case: average the observed Y. (ii) Model-based: fit the regression $Y \approx b_0 + b_1 X$ on the complete cases, then average the prediction $b_0 + b_1 X$ over *all* n values of X; this is the maximum-likelihood estimate of $\mathbb{E}[Y]$ under a linear-Gaussian model for $Y \mid X$.

	fitted (b_0, b_1)	complete-case mean	model-based mean
MCAR	(1.003, 0.995)	1.006	1.005
MAR	(0.995, 0.996)	0.467	0.997
MNAR	(0.463, 0.763)	0.127	0.464

Under MAR, records with large X (hence large Y) are deleted more often, so the complete-case mean is too low; but $R \perp\!\!\!\perp Y \mid X$, so the regression of Y on X is still unbiased on the complete cases and averaging it over all X (whose mean is 0.002) recovers $\mathbb{E}[Y]$. Under MNAR the deletion depends on Y itself even given X, so the regression fitted on complete cases is already biased ($b_0 = 0.463$, $b_1 = 0.763$ instead of $1, 1$) and no analysis of the observed data alone repairs it.

PYTHON LAB · `s10_examples.py` · `ex_missing_mechanisms()`

```
MCAR missing=0.399 complete-case mean=1.006 model-based mean=1.005;  MAR
missing=0.500 0.467 0.997; MNAR missing=0.499 0.127 0.464.
```

Table 10.2. Missing-data mechanisms in the logs of an AI agent.

	Definition	Agent-log example	Complete cases	Likelihood / EM
MCAR	$R \perp\!\!\!\perp \mathbf{Y}$	server drops random records	unbiased	valid
MAR	$R \perp\!\!\!\perp \mathbf{Y}_{\text{mis}} \mid \mathbf{Y}_{\text{obs}}$	mobile users skip the rating	biased	valid
MNAR	R depends on $\mathbf{Y}_{\text{mis}}$	unhappy users never rate	biased	biased unless R modelled

PART III · LATENT VARIABLES

10.7 When Observability Fails: Latent Variables

Now suppose some variables are never recorded. A marketing team observes purchase amounts but never directly asks customers which underlying "segment" (budget-conscious, aspirational, luxury) they belong to; a language model sees words but never sees the latent topic that generated them. Call the observed variables X and the latent (hidden) variables C; Fig. 10.5 shows this structure using *plate notation*, a compact way to draw "repeat this substructure N times." A latent variable is the extreme case of missing data: it is missing in every record, so $R \equiv 1$ does not depend on any value; the mechanism is trivially MCAR (hence MAR), and Theorem 10.5 lets us work with $p(\mathbf{y}_{\text{obs}} \mid \theta)$ alone.

The problem: Proposition 10.1's decomposition assumed we could plug $\mathbf{x}^{(j)}$ directly into every

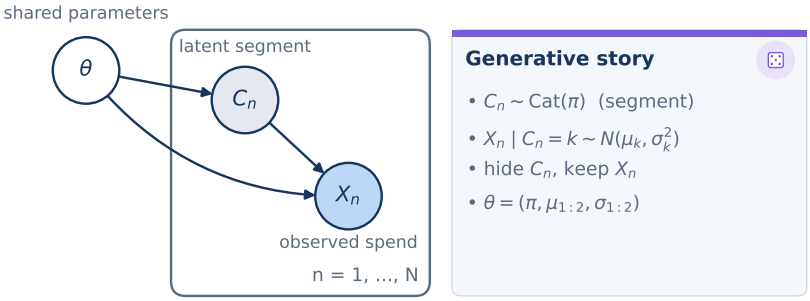


Figure 10.5. Plate notation: the box means "one copy of $C_n \to X_n$ for each of $n = 1, \ldots, N$ customers," with the parameters $\theta = (\pi, \mu_{1:2}, \sigma_{1:2})$ (white node, outside the plate) shared across all copies. The blue node X_n (spend) is observed; the grey node C_n (segment) is latent.

factor. With C unobserved, the likelihood of the data we *can* see requires summing C out:

$$\ell(\theta; \mathcal{D}) = \sum_{j=1}^{N} \log P(x^{(j)}; \theta) = \sum_{j=1}^{N} \log \sum_{c} P(x^{(j)}, c; \theta),$$

and the log of a *sum* no longer decomposes into separate per-node terms the way Proposition 10.1 did. This is not a minor inconvenience — it usually makes direct maximization analytically intractable.

PART III · LATENT VARIABLES

10.8 The EM Algorithm

The Expectation-Maximization (EM) algorithm (Dempster, Laird and Rubin, 1977) sidesteps this by alternating between (1) using the *current* parameter estimate to infer a distribution over the missing C given what is observed — exactly a Session-3 inference query — and (2) treating those inferred (soft) values as if they were observed, reducing back to the easy, decomposable MLE problem of Part I.

Algorithm 10.1 Expectation–Maximization

1: Initialize $\theta^{(0)}$ arbitrarily (e.g., randomly).
2: **for** $t = 0, 1, 2, \ldots$ until convergence **do**
3: **E-step:** for each data point j, compute $q_j(c) := P(c \mid x^{(j)}; \theta^{(t)})$ (an inference query — Session 7's VE).
4: **M-step:** set $\theta^{(t+1)} = \arg\max_\theta \sum_j \sum_c q_j(c) \log P(x^{(j)}, c; \theta)$ (an ordinary weighted MLE, decomposable exactly as in Proposition 10.1, now with "soft counts" $q_j(c)$ replacing observed indicator values).
5: **end for**

Proposition 10.6 (ELBO decomposition)**.** *For any distribution q over c and any θ,* $\log P(x; \theta) = \underbrace{\sum_c q(c) \log \frac{P(x,c;\theta)}{q(c)}}_{\mathcal{L}(q,\theta)\ \textit{(ELBO)}} + \mathrm{KL}\big(q \,\|\, P(\cdot \mid x; \theta)\big)$.

Proof. Write $P(x, c; \theta) = P(x; \theta) P(c \mid x; \theta)$ inside $\mathcal{L}$: $\mathcal{L}(q, \theta) = \log P(x; \theta) - \sum_c q(c) \log \frac{q(c)}{P(c|x;\theta)}$. □

For a dataset, apply Proposition 10.6 to each record with its own q_j and sum: $\ell(\theta;\mathcal{D}) = \mathcal{L}(q,\theta) + \sum_j \mathrm{KL}\big(q_j \,\|\, P(\cdot \mid x^{(j)};\theta)\big)$ with $\mathcal{L}(q,\theta) = \sum_j \sum_c q_j(c) \log \frac{P(x^{(j)},c;\theta)}{q_j(c)}$. Splitting the log,

$$\mathcal{L}(q,\theta) = Q(\theta \mid \theta^{(t)}) + H(q), \qquad Q(\theta \mid \theta^{(t)}) = \sum_j \sum_c q_j(c) \log P(x^{(j)}, c;\theta), \tag{10.2}$$

with the entropy term $H(q) = -\sum_j \sum_c q_j(c) \log q_j(c)$, where Q (the *expected complete-data log-likelihood*) is evaluated with $q_j = P(\cdot \mid x^{(j)};\theta^{(t)})$. EM is *coordinate ascent on the ELBO*: the E-step maximizes $\mathcal{L}$ over q by setting $q_j = P(\cdot \mid x^{(j)};\theta^{(t)})$, which makes every KL term zero; the M-step maximizes $\mathcal{L}$ over θ, and since $H(q)$ does not involve θ this is exactly maximizing $Q(\theta \mid \theta^{(t)})$, the M-step of Algorithm 10.1.

Theorem 10.6 (EM monotonicity). *The sequence of observed-data log-likelihoods produced by EM is non-decreasing: $\ell(\theta^{(t+1)};\mathcal{D}) \geq \ell(\theta^{(t)};\mathcal{D})$ for every t.*

The proof (Section 10.20) uses the decomposition (10.2) and $\mathrm{KL} \geq 0$ (Gibbs' inequality, itself Jensen's inequality for the concave log) to show the M-step maximizes a lower bound on ℓ that is guaranteed to touch ℓ exactly at $\theta^{(t)}$ — so any improvement to the bound is a genuine (or at worst equal) improvement to the true log-likelihood. EM is not guaranteed to reach a global maximum (it can and does get stuck in local optima, so multiple random restarts are standard practice), but Theorem 10.6 guarantees it never gets *worse*, which is the essential property that makes it a reliable, widely-used algorithm rather than a heuristic.

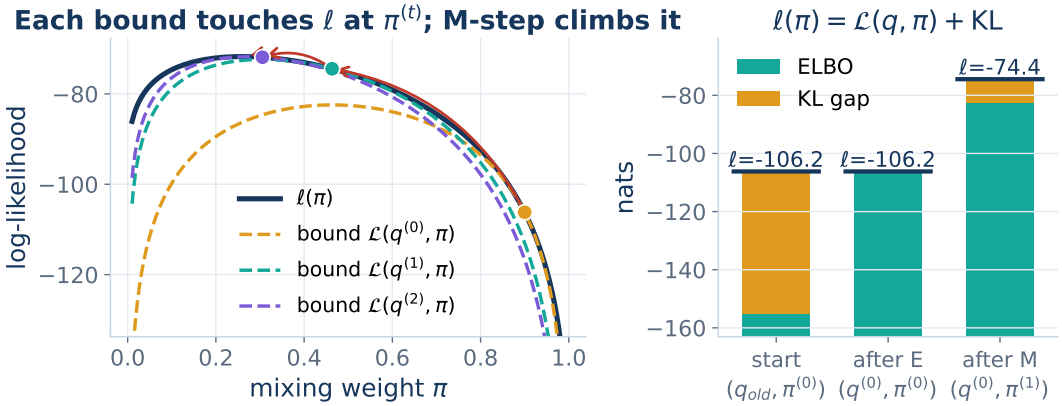


Figure 10.6. EM as minorize–maximize, computed exactly for an illustrative mixture $\pi N(3,1)+(1-\pi)N(0,1)$ in which only the weight π is unknown (40 points simulated with $\pi = 0.3$, seed 17, `fig_em_bound()` in `s10_figures.py`); EM starts at $\pi^{(0)} = 0.9$. Left: each dashed bound $\mathcal{L}(q^{(t)},\pi)$ lies below $\ell(\pi)$ and touches it at $\pi^{(t)}$; maximizing it moves to $\pi^{(t+1)}$ (three iterations shown). Right: starting from an arbitrary q_{old} (every responsibility 0.5), the E-step closes the KL gap without changing ℓ; the M-step raises the ELBO and therefore ℓ (the values of ℓ are printed above the bars).

PART III · LATENT VARIABLES

10.9 EM for Gaussian Mixtures in Full

Proposition 10.7 (EM updates for a Gaussian mixture). *For $p(x) = \sum_{k=1}^{K} \pi_k N(x;\mu_k,\sigma_k^2)$,*

one EM iteration is

$$E{:}\ r_{jk} = \frac{\pi_k N(x_j;\mu_k,\sigma_k^2)}{\sum_l \pi_l N(x_j;\mu_l,\sigma_l^2)}; \quad M{:}\ \pi_k = \frac{N_k}{N},\ \ \mu_k = \frac{\sum_j r_{jk}x_j}{N_k},\ \ \sigma_k^2 = \frac{\sum_j r_{jk}(x_j-\mu_k)^2}{N_k},$$

with $N_k = \sum_j r_{jk}$. *In* d *dimensions,* $\Sigma_k = \frac{1}{N_k}\sum_j r_{jk}(\mathbf{x}_j-\boldsymbol{\mu}_k)(\mathbf{x}_j-\boldsymbol{\mu}_k)^\top$.

Proof. $Q(\theta) = \sum_j \sum_k r_{jk}\big[\log\pi_k - \frac{1}{2}\log(2\pi\sigma_k^2) - \frac{(x_j-\mu_k)^2}{2\sigma_k^2}\big]$. The π-part $\sum_k N_k \log\pi_k$ is maximized on the simplex at $\pi_k = N_k/N$ (Theorem 10.1). $\partial Q/\partial\mu_k = \sum_j r_{jk}(x_j-\mu_k)/\sigma_k^2 = 0$ gives the weighted mean. $\partial Q/\partial\sigma_k^2 = \sum_j r_{jk}\big[-\frac{1}{2\sigma_k^2} + \frac{(x_j-\mu_k)^2}{2\sigma_k^4}\big] = 0$ gives the weighted variance (evaluated at the new μ_k). These stationary points are maxima: Q is a concave quadratic in μ_k, and as a function of $v=\sigma_k^2$ the k-th term is $-\frac{N_k}{2}\log v - \frac{S_k}{2v}$ with $S_k = \sum_j r_{jk}(x_j-\mu_k)^2 > 0$, which tends to $-\infty$ as $v\to 0$ and as $v\to\infty$ and has the single stationary point $v = S_k/N_k$. □

Worked Example 10.8 (One full EM iteration with weights, means and variances)**.** Data $x = (1, 1.5, 2, 6, 7, 7.5)$; components $k = 0, 1$ started at $\pi = (0.5, 0.5)$, $\mu = (1, 7)$, $\sigma^2 = (1, 1)$; write $\varphi(z) = e^{-z^2/2}/\sqrt{2\pi}$.

1. *E-step.* With equal weights and variances, $r_{j0} = e^{-(x_j-1)^2/2}/\big(e^{-(x_j-1)^2/2} + e^{-(x_j-7)^2/2}\big)$. The least certain points are $x=2$: $r_{j0} = e^{-0.5}/(e^{-0.5}+e^{-12.5}) = 0.999994$, and $x=6$: $r_{j0} = e^{-12.5}/(e^{-12.5}+e^{-0.5}) = 0.000006$. So, to four decimals, $r_{\cdot 0} = (1,1,1,0,0,0)$ and $N_k = (3,3)$.
2. *M-step.* $\pi = (3/6, 3/6) = (0.5, 0.5)$; $\mu_0 = (1+1.5+2)/3 = 1.5$, $\mu_1 = (6+7+7.5)/3 = 20.5/3 = 6.8333$; $\sigma_0^2 = \big((-0.5)^2 + 0^2 + 0.5^2\big)/3 = 0.5/3 = 0.1667$, $\sigma_1^2 = \big((6-6.8333)^2 + (7-6.8333)^2 + (7.5-6.8333)^2\big)/3 = (0.6944 + 0.0278 + 0.4445)/3 = 1.1667/3 = 0.3889$.
3. *Log-likelihood* $\ell = \sum_j \log\sum_k \pi_k N(x_j;\mu_k,\sigma_k^2)$. Before: the other component contributes at most $e^{-12.5}$, so $\log p(x_j) \approx \log\big(0.5\,\varphi(d_j)\big)$ with distance $d_j = 0, 0.5, 1$ to the nearest mean, i.e. $-1.61209, -1.73709, -2.11208$, each occurring twice: $\ell^{(0)} = 2(-1.61209 - 1.73709 - 2.11208) = -10.9225$. After the M-step the per-point terms are $(-1.46617, -0.71633, -1.46612, -2.03264, -1.17563, -1.71129)$, summing to $\ell^{(1)} = -8.5682$.

PYTHON LAB · `s10_examples.py` · `ex_gmm_full_iteration()`

```
r(c=0|x=2) = 0.999994, r(c=0|x=6) = 0.000006;   pi^(1) = [0.5 0.5], mu^(1) =
[1.5 6.8333], var^(1) = [0.1667 0.3889]; log-lik:  -10.9225 -> -8.5682.
```

COMMON PITFALL The MLE of a variance is biased, $\mathbb{E}\hat\sigma^2 = \frac{n-1}{n}\sigma^2$ for n i.i.d. Gaussian observations (and $\mathbb{E}[\mathrm{RSS}/N] = \frac{N-p}{N}\sigma^2$ in Worked Example 10.2, with p the number of regression coefficients); worse, the Gaussian-mixture likelihood is *unbounded*: put $\mu_k = x_j$ and let $\sigma_k \to 0$; then the term $\pi_k N(x_j; x_j, \sigma_k^2) = \pi_k/(\sqrt{2\pi}\sigma_k) \to \infty$ while the other points keep positive density from the other components, so $\ell \to +\infty$. EM can collapse a component onto one point. In practice: floor the variances, add a weak inverse-gamma (1-D) or inverse-Wishart (d-D) prior and run MAP-EM, or restart.

A 2-D illustration (simulation, `ex_gmm_2d()`). Illustrative data: 450 points from three bivariate Gaussian clusters of sizes $180, 150, 120$ (true weights $0.4, 0.333, 0.267$; means $(0,0)$, $(4,1)$, $(1.5,4)$;

covariances listed in `gmm_2d_data()`, seed 7). EM for a three-component full-covariance mixture is deliberately started badly: the three means are random points among the 60 left-most data points, every covariance equals the overall sample covariance, and the weights are equal. The log-likelihood still climbs monotonically, $-2399 \to -1799 \to -1733 \to -1555$ at iterations $0, 1, 5, 40$, and the final weights $(0.395, 0.267, 0.338)$ match the true $(0.4, 0.333, 0.267)$ up to relabeling of the components.

PYTHON LAB · s10_examples.py · ex_gmm_2d()

```
N = 450; log-lik at iterations 0, 1, 5, 40:  -2399, -1799, -1733, -1555;
monotone:  True; final weights:  [0.395 0.267 0.338].
```

PART III · LATENT VARIABLES

10.10 EM for Bayesian Networks: Expected Counts

For a BN with some entries missing in some records, the M-step needs the counts $N[x, u]$, which are not observed. Replace them by *expected sufficient statistics*

$$\bar{M}_t[x,u] = \sum_{j=1}^{N} P\big(X_i{=}x,\ \mathrm{pa}_i{=}u \mid \mathbf{o}^{(j)}; \theta^{(t)}\big), \qquad \theta^{(t+1)}_{x|u} = \frac{\bar{M}_t[x,u]}{\sum_{x'} \bar{M}_t[x',u]}, \tag{10.3}$$

where $\mathbf{o}^{(j)}$ is the observed part of record j. A record in which the whole family is observed contributes the indicator $\mathbb{1}\{x_i^{(j)}{=}x,\ \mathrm{pa}_i^{(j)}{=}u\}$, exactly as in (10.1); an incomplete record contributes a *family marginal* computed by inference (Session 7: one clique-tree calibration per record gives all family marginals at once). With a $\mathrm{Dir}(\boldsymbol{\alpha}_u)$ prior on each row, MAP-EM uses the MAP formula of Theorem 10.3 with expected counts, $\theta^{(t+1)}_{x|u} = (\bar{M}_t[x,u] + \alpha_{x|u} - 1)/\sum_{x'}(\bar{M}_t[x',u] + \alpha_{x'|u} - 1)$; with $\alpha_{x|u} = 2$ this is "add one to every expected count".

Worked Example 10.9 (EM with expected counts in $A \to B$)**.** A hypothetical log of $N = 140$ records for binary $A \to B$, with parameters $\theta_A = P(A{=}1)$, $\theta_{B|1} = P(B{=}1 \mid A{=}1)$, $\theta_{B|0} = P(B{=}1 \mid A{=}0)$:

- 80 complete records: $(A,B) = (1,1){\times}30$, $(1,0){\times}10$, $(0,1){\times}15$, $(0,0){\times}25$;
- 40 records with B missing: $A{=}1$ in 20, $A{=}0$ in 20;
- 20 records with A missing: $B{=}1$ in 12, $B{=}0$ in 8.

Start at $\theta^{(0)}$: $\theta_A = \theta_{B|1} = \theta_{B|0} = 0.5$.
Observed-data log-likelihood. A complete record contributes $\log P(a,b)$, a B-missing record $\log P(a)$, an A-missing record $\log P(b) = \log \sum_a P(a)P(b \mid a)$. At $\theta^{(0)}$ these are $\log 0.25$, $\log 0.5$ and $\log(0.5 \cdot 0.5 + 0.5 \cdot 0.5) = \log 0.5$, so $\ell^{(0)} = 80 \log 0.25 + 40 \log 0.5 + 20 \log 0.5 = -110.904 - 27.726 - 13.863 = -152.492$.
E-step. For an A-missing record, $P(A{=}1 \mid B{=}b) = \frac{\theta_A P(b|A=1)}{\theta_A P(b|A=1) + (1-\theta_A)P(b|A=0)} = \frac{0.5 \cdot 0.5}{0.5 \cdot 0.5 + 0.5 \cdot 0.5} = 0.5$ for both b, so each such record is split evenly between $A{=}1$ and $A{=}0$. For a B-missing record, $P(B{=}1 \mid A{=}a) = \theta_{B|a} = 0.5$. Expected counts, term by term (complete + B-missing

+ A-missing):

$$\bar{M}[A{=}1] = (30+10) + 20 + (0.5\cdot 12 + 0.5\cdot 8) = 40+20+10 = 70,$$
$$\bar{M}[A{=}0] = (15+25) + 20 + (0.5\cdot 12 + 0.5\cdot 8) = 40+20+10 = 70,$$
$$\bar{M}[A{=}1, B{=}1] = 30 + 0.5\cdot 20 + 0.5\cdot 12 = 30+10+6 = 46,$$
$$\bar{M}[A{=}0, B{=}1] = 15 + 0.5\cdot 20 + 0.5\cdot 12 = 15+10+6 = 31.$$

M-step. $\theta_A = 70/140 = 0.5$, $\theta_{B|1} = 46/70 = 0.6571$, $\theta_{B|0} = 31/70 = 0.4429$; now $\ell^{(1)} = -146.881$.
Iterating 200 times (to convergence) gives $\hat{\theta}_A = 0.5019$, $\hat{\theta}_{B|1} = 0.7557$, $\hat{\theta}_{B|0} = 0.3823$, with ℓ rising monotonically from -152.492 to -145.640. The complete-case estimates $30/40 = 0.750$ and $15/40 = 0.375$ differ because the A-missing records carry information about $P(B \mid A)$ (the B-missing records, being MAR given A, only inform θ_A).

PYTHON LAB · s10_examples.py · ex_em_missing_cpt()

```
l(theta^(0)) = -110.904 + -27.726 + -13.863 = -152.492;      expected counts
M[A=1]=70.00, M[A=1,B=1]=46.00, M[A=0,B=1]=31.00;  after 1 iteration:  theta
= [0.5 0.6571 0.4429];   converged (200 iterations):  0.5019, 0.7557, 0.3823;
log-lik:  -152.492 -> -146.881 -> ...  -> -145.640; monotone:  True.
```

PART III · LATENT VARIABLES

10.11 Hard EM versus Soft EM

Hard EM replaces the posterior q_j by a point mass on its mode $\hat{c}_j = \arg\max_c P(c \mid x^{(j)}; \theta^{(t)})$ and then refits θ on the completed data. For a mixture of spherical Gaussians with equal, known variances and weights, the mode is the nearest mean, so hard EM is exactly k-means. Hard EM maximizes the joint $\max_{c_{1:N},\theta} \log P(\mathbf{x}, \mathbf{c}; \theta)$, a *different* objective from $\ell(\theta)$: it needs only (cheaper) MAP inference, but its estimates are biased, and stay biased as $N \to \infty$ (inconsistent), when components overlap.

Simulation (`ex_hard_vs_soft()`, seed 21): 4000 points from $\frac{1}{2}N(0,1) + \frac{1}{2}N(2,1)$; $\sigma = 1$ and the weights $\frac{1}{2}$ are known, only the two means are learned, both algorithms start at $(-1, 3)$ and run 200 iterations. Soft EM returns $(-0.033, 2.015)$, close to the truth $(0, 2)$; hard EM returns $(-0.177, 2.191)$. Hard assignment splits the data at the midpoint between the means, so each cluster mean is computed from a truncated sample (the upper tail of $N(0,1)$ is given to the other cluster and vice versa), which pushes the two means apart.

PYTHON LAB · s10_examples.py · ex_hard_vs_soft()

```
true means (0, 2); soft EM -> [-0.033 2.015]; hard EM -> [-0.177 2.191].
```

PART III · LATENT VARIABLES

10.12 Worked Examples: Learning by Hand

Worked Example 10.10 (EM by hand: one iteration on a tiny mixture)**.** Two data points $x^{(1)} = 2$, $x^{(2)} = 7$ are believed to come from a two-component Gaussian mixture with known equal variance $\sigma^2 = 1$ and known mixing weights $\pi_0 = \pi_1 = 0.5$, but unknown means, initialized at $\mu_0^{(0)} = 3$, $\mu_1^{(0)} = 6$. *E-step for $x^{(1)} = 2$:* the responsibility is $q_1(c) = \pi_c N(2; \mu_c, 1) / \sum_{c'} \pi_{c'} N(2; \mu_{c'}, 1)$; the common factors $\pi_c = 0.5$ and $1/\sqrt{2\pi}$ cancel, leaving the unnormalized values $\exp(-\frac{1}{2}(2-3)^2) = e^{-0.5} = 0.6065$ for component 0 and $\exp(-\frac{1}{2}(2-6)^2) = e^{-8} = 0.00034$ for component 1. Normalizing, $q_1(c{=}0) = 0.6065/(0.6065 + 0.00034) = 0.9994$ and $q_1(c{=}1) = 0.0006$ — point 1 is confidently assigned to component 0. The symmetric computation for $x^{(2)} = 7$ (distances 4 and 1) gives $q_2(c{=}0) = 0.0006$, $q_2(c{=}1) = 0.9994$. *M-step:* the new mean for component 0 is the responsibility-weighted average

$$\mu_0^{(1)} = \frac{q_1(0) \cdot 2 + q_2(0) \cdot 7}{q_1(0) + q_2(0)} = \frac{0.9994 \cdot 2 + 0.0006 \cdot 7}{0.9994 + 0.0006} = 2.003,$$

essentially the point $x^{(1)} = 2$ itself, because point 2 has almost no weight in component 0; similarly $\mu_1^{(1)} = (0.0006 \cdot 2 + 0.9994 \cdot 7)/1 = 6.997$. From a mediocre initialization, one iteration moves each mean almost onto "its" point, because the points are far apart relative to $\sigma = 1$.

PYTHON LAB · s10_examples.py · ex_em_tiny()

```
unnormalised for x=2:  0.6065, 0.00034;    q(c=0|x=2) = 0.9994, q(c=1|x=7) =
0.9994; mu0^(1) = 2.003, mu1^(1) = 6.997.
```

Worked Example 10.11 (Customer segmentation via EM)**.** *Hypothetical (simulated) data* (`customer_data()`, seed 42): a firm has $N = 400$ customers' purchase amounts generated from two latent segments, $\pi = (0.35, 0.65)$, $\mu = (2, 7)$, $\sigma = (1, 1.4)$; the segment label is then discarded. *EM* (`em_gmm_1d()`) fits a two-component 1-D Gaussian mixture with the updates of Proposition 10.7, started at $\pi = (0.5, 0.5)$, means equal to two randomly chosen data points (0.778 and 5.600), and both standard deviations equal to the standard deviation of all the data (2.732); it runs 25 iterations. *Result.* The log-likelihood rises from -1046.83 to -882.73, increasing at every step exactly as Theorem 10.6 guarantees (Fig. 10.7), and the fit is $\hat{\pi} = (0.36, 0.64)$, $\hat{\mu} = (1.928, 7.042)$, $\hat{\sigma} = (1.064, 1.266)$, close to the generating values (Fig. 10.8). EM was only told that there are two components, never which customer belongs to which; it recovered the mixing proportions and both per-segment spending distributions from the shape of the unlabeled data alone.

PYTHON LAB · s10_examples.py · ex_customer_segmentation()

```
initialisation:  mu = [0.778 5.6 ], sigma = 2.732;      EM pi=[0.36 0.64],
mu=[1.928 7.042], sigma=[1.064 1.266];     log-lik:  start -1046.83 -> end
-882.73; monotone:  True.
```

Worked Example 10.12 (Why the E-step is "just" Session 7's inference)**.** Suppose the mixture above is instead written as a two-node Bayesian network $C \to X$ (segment causes

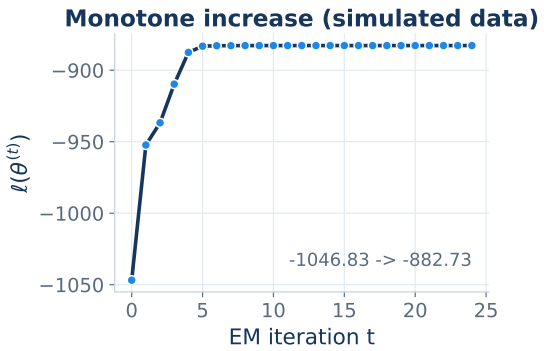


Figure 10.7. EM on the simulated customer data of Worked Example 10.11: $\ell(\theta^{(t)})$ for $t = 0, \ldots, 24$ increases monotonically, from -1046.83 to -882.73.

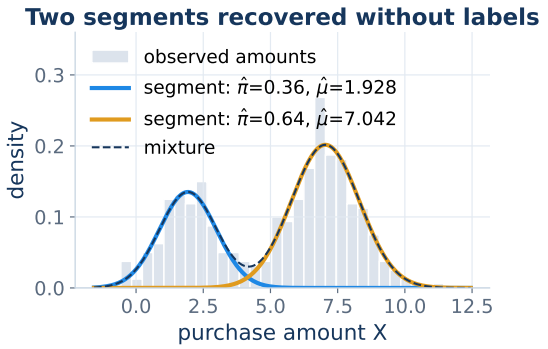


Figure 10.8. Recovered two-segment mixture (weighted components $\hat{\pi}_k N(x; \hat{\mu}_k, \hat{\sigma}_k^2)$ and their sum) over the histogram of the 400 unlabeled purchase amounts.

purchase behavior), with $P(C)$ the mixing weights and $P(X \mid C)$ the per-segment spending distribution — exactly the plate diagram in Fig. 10.5 unrolled for one customer. The E-step's $q_j(c) = P(c \mid x^{(j)}; \theta^{(t)})$ is then *literally* the query "$P(C \mid X{=}x^{(j)})$" from a two-node network with current parameters $\theta^{(t)}$ — solvable in one line by Bayes' rule, $q_j(c) = \pi_c^{(t)} N(x^{(j)}; \mu_c^{(t)}, \sigma_c^{2(t)}) / \sum_{c'} \pi_{c'}^{(t)} N(x^{(j)}; \mu_{c'}^{(t)}, \sigma_{c'}^{2(t)})$. This is a trivial case of Session 7's variable elimination: there is one query variable (C), one evidence variable (X) and nothing to eliminate. For larger latent-variable networks, the E-step is exactly VE with the current $\theta^{(t)}$ plugged in; nothing new needs to be learned computationally, only combined with the M-step's ordinary weighted MLE.

Additional Worked Examples (Instructor Supplement)

Worked Example 10.13 (A second fully-observed counting example)**.** A two-node network $A \to B$ has $N = 10$ fully-observed records: $(A, B) = (1,1) \times 4,\ (1,0) \times 1,\ (0,1) \times 2,\ (0,0) \times 3$. Compute the full MLE CPTs.

Solution. $\hat{P}(A{=}1) = 5/10 = 0.5$ (five records have $A{=}1$: the four $(1,1)$'s and one $(1,0)$). Among those five, $B{=}1$ occurs 4 times, so $\hat{P}(B{=}1 \mid A{=}1) = 4/5 = 0.8$. Among the five records with $A{=}0$, $B{=}1$ occurs 2 times, so $\hat{P}(B{=}1 \mid A{=}0) = 2/5 = 0.4$. Exactly as Proposition 10.1 guarantees, each CPT row was estimated purely by counting within its own parent configuration, with no reference to the other rows at all.

PYTHON LAB · `s10_examples.py` · `ex_counting_two_node()`

```
P_hat(A=1) = 5/10 = 0.5,  P_hat(B=1 | A=1) = 4/5 = 0.8,  P_hat(B=1 | A=0) = 2/5
= 0.4.
```

Worked Example 10.14 (Deriving the Bernoulli MLE by calculus)**.** Theorem 10.1 derived the counting formula with Gibbs' inequality or a Lagrange multiplier; here is the plain-calculus version for the simplest case, a single Bernoulli(θ) node with i.i.d. data $x_1, \ldots, x_N \in \{0, 1\}$, where the constraint is eliminated by writing $P(X{=}0) = 1 - \theta$.

Solution. $\ell(\theta) = \sum_{i=1}^{N} \big[x_i \log\theta + (1-x_i)\log(1-\theta)\big]$. Differentiating and setting to zero:

$$\frac{d\ell}{d\theta} = \sum_i \left[\frac{x_i}{\theta} - \frac{1-x_i}{1-\theta}\right] = 0 \implies (1-\theta)\sum_i x_i = \theta\sum_i (1-x_i) = \theta N - \theta\sum_i x_i,$$

$$\implies \sum_i x_i - \theta\sum_i x_i = \theta N - \theta\sum_i x_i \implies \sum_i x_i = \theta N \implies \hat\theta = \frac{1}{N}\sum_i x_i,$$

the sample mean — exactly the empirical frequency used in every counting formula in this session, now derived from first-order calculus rather than merely asserted. The second derivative $\ell''(\theta) = -\sum_i x_i/\theta^2 - \sum_i (1-x_i)/(1-\theta)^2$ is negative for every $0 < \theta < 1$, so ℓ is strictly concave and this critical point is the maximum. *Numerical check:* for $x = (1,0,1,1,0,1,1,0,1,1)$ (7 ones in $N = 10$), $\hat\theta = 0.7$ and $\ell(\hat\theta) = 7\log 0.7 + 3\log 0.3 = -6.1086$; maximizing ℓ over a fine grid of θ values returns the same 0.700.

PYTHON LAB · `s10_examples.py` · `ex_bernoulli_mle()`

```
sample mean = 0.700;        grid argmax of log-likelihood = 0.700;        max
log-likelihood = -6.1086.
```

Worked Example 10.15 (EM with a genuinely ambiguous point)**.** Three points $x = (1, 5, 9)$ come from a two-component mixture, $\sigma^2 = 1$, mixing weights 0.5/0.5, initialized at $\mu_0^{(0)} = 2, \mu_1^{(0)} = 8$. Unlike Worked Example 10.10 (where both points were confidently assigned), $x = 5$ sits exactly midway.

E-step (as in Worked Example 10.10, the equal weights and the factor $1/\sqrt{2\pi}$ cancel). For x=1: unnormalized responsibilities $e^{-\frac{1}{2}(1-2)^2} = e^{-0.5} = 0.6065$ (component 0) vs. $e^{-\frac{1}{2}(1-8)^2} = e^{-24.5} = 2.3\times10^{-11}$ (component 1), so $q(0 \mid 1) = 1$ to four decimals. Symmetrically $q(1 \mid 9) \approx 1$. For x=5: $e^{-\frac{1}{2}(5-2)^2} = e^{-4.5} = 0.0111$ for *both* components (since $|5-2| = |5-8| = 3$), so the responsibilities are exactly **tied**: $q(0 \mid 5) = q(1 \mid 5) = 0.5$.

M-step. $\mu_0^{(1)} = \dfrac{q(0\mid1)(1) + q(0\mid5)(5) + q(0\mid9)(9)}{q(0\mid1) + q(0\mid5) + q(0\mid9)} = \dfrac{1(1) + 0.5(5) + 0(9)}{1 + 0.5 + 0} = \dfrac{3.5}{1.5} \approx \mathbf{2.333}$,

and by symmetry $\mu_1^{(1)} = \dfrac{0(1) + 0.5(5) + 1(9)}{0 + 0.5 + 1} = \dfrac{11.5}{1.5} \approx \mathbf{7.667}$. The ambiguous point $x = 5$ pulls *both* means slightly toward the center (from $2 \to 2.333$ and $8 \to 7.667$) by contributing a genuine half-share to each — this is exactly what "soft" assignment means in practice: EM does not have to commit an uncertain point to one cluster, unlike a hard-assignment algorithm (e.g., k-means), which would be forced to break the tie arbitrarily.

PYTHON LAB · `s10_examples.py` · `ex_em_ambiguous()`

```
unnormalised:  x=1:  0.6065, 2.3e-11; x=5:  0.0111, 0.0111;    q(c=0|x) = [1.
0.5 0.];  mu0^(1) = 3.5/1.5, mu1^(1) = 11.5/1.5;  mu0^(1) = 2.333, mu1^(1) =
7.667.
```

PART IV

Mathematical Statistics of Learning

How good are our estimates, and how fast does EM converge?

PART IV · STATISTICS

10.13 Score, Fisher Information and the Cramér–Rao Bound

Definition 10.4 (Score and Fisher information)**.** For a regular model $p(x;\theta)$, the *score* is $s(\theta;x)=\nabla_\theta \log p(x;\theta)$ and the *Fisher information* of one observation is $I_1(\theta)=\mathbb{E}_\theta[ss^\top]=-\mathbb{E}_\theta[\nabla^2_\theta \log p(X;\theta)]$; for n i.i.d. observations $I_n=nI_1$. The *observed information* is $J_n(\hat\theta)=-\nabla^2\ell(\hat\theta)$, the curvature of the log-likelihood at its maximum: the more data, the sharper the peak.

Theorem 10.7 (Cramér–Rao lower bound)**.** *Let $T(X_{1:n})$ be an unbiased estimator of $g(\theta)$, $\theta\in\mathbb{R}$, in a model where the support does not depend on θ and differentiation under the integral sign is allowed. Then* $\operatorname{Var}_\theta(T)\ge g'(\theta)^2/I_n(\theta)$.

Proof. Let $S=\sum_i s(\theta;X_i)$. First, $\mathbb{E}_\theta S=\int \partial_\theta p_\theta=\partial_\theta\int p_\theta=0$. Second,

$$g'(\theta)=\partial_\theta\int T p_\theta=\int T\,(\partial_\theta\log p_\theta)\,p_\theta=\mathbb{E}_\theta[TS]=\operatorname{Cov}_\theta(T,S).$$

By Cauchy–Schwarz, $g'(\theta)^2\le \operatorname{Var}(T)\operatorname{Var}(S)=\operatorname{Var}(T)\,I_n(\theta)$, since $\operatorname{Var}(S)=nI_1(\theta)$ by independence. □

Worked Example 10.16 (Fisher information of Bernoulli and categorical CPDs (analytical))**.** *Bernoulli:* $\log p=x\log\theta+(1-x)\log(1-\theta)$, $-\partial^2_\theta\log p=x/\theta^2+(1-x)/(1-\theta)^2$, and taking the expectation with $\mathbb{E}X=\theta$ gives $I_1(\theta)=\theta/\theta^2+(1-\theta)/(1-\theta)^2=1/\theta+1/(1-\theta)=1/(\theta(1-\theta))$. The MLE k/n is unbiased with variance $\theta(1-\theta)/n=1/I_n(\theta)$: it attains the bound of Theorem 10.7 (with $g(\theta)=\theta$), i.e. it is *efficient*. At $\theta=0.3$: $I_1=1/(0.3\cdot0.7)=4.7619$ and, for $n=50$, the bound is $0.21/50=0.00420$; the variance of k/n over 200 000 simulated samples of size 50 (seed 3) is also 0.00420.
Categorical with free coordinates $(\theta_1,\dots,\theta_{K-1})$, $\theta_K=1-\sum_{k<K}\theta_k$, so $\log p(x)=\sum_{k<K}\mathbb{1}\{x{=}k\}\log\theta_k+\mathbb{1}\{x{=}K\}\log\theta_K$: $-\partial^2\log p/\partial\theta_k\partial\theta_l=\mathbb{1}\{x{=}k\}\delta_{kl}/\theta_k^2+\mathbb{1}\{x{=}K\}/\theta_K^2$. Taking expectations ($\mathbb{E}\mathbb{1}\{X{=}k\}=\theta_k$) gives $I_1=D+\theta_K^{-1}\mathbf{1}\mathbf{1}^\top$ with $D=\operatorname{diag}(1/\theta_1,\dots,1/\theta_{K-1})$. Sherman–Morrison, $(D+c\mathbf{1}\mathbf{1}^\top)^{-1}=D^{-1}-\frac{cD^{-1}\mathbf{1}\mathbf{1}^\top D^{-1}}{1+c\,\mathbf{1}^\top D^{-1}\mathbf{1}}$, with $c=1/\theta_K$, $D^{-1}\mathbf{1}=\boldsymbol{\theta}=(\theta_1,\dots,\theta_{K-1})^\top$ and $1+c\,\mathbf{1}^\top D^{-1}\mathbf{1}=1+(1-\theta_K)/\theta_K=1/\theta_K$, gives $I_1^{-1}=\operatorname{diag}(\boldsymbol{\theta})-\boldsymbol{\theta}\boldsymbol{\theta}^\top$, the multinomial covariance. For $\theta=(0.5,0.3,0.2)$: $I_1=\left(\begin{smallmatrix}2{+}5 & 5\\ 5 & 3.3333{+}5\end{smallmatrix}\right)=\left(\begin{smallmatrix}7 & 5\\ 5 & 8.3333\end{smallmatrix}\right)$ and $I_1^{-1}=\left(\begin{smallmatrix}0.5{-}0.25 & -0.15\\ -0.15 & 0.3{-}0.09\end{smallmatrix}\right)=\left(\begin{smallmatrix}0.25 & -0.15\\ -0.15 & 0.21\end{smallmatrix}\right)$.

PYTHON LAB · `s10_examples.py` · `ex_fisher_cr()`

```
I_1(theta) = 1/(0.3*0.7) = 4.7619, CR bound = 0.21/50 = 0.00420;   simulated
Var(MLE) over R=200000 samples (seed 3) = 0.00420;  I_1 = [[7 5] [5 8.3333]],
I_1^-1 = [[0.25 -0.15] [-0.15 0.21]].
```

MATHEMATICAL STATISTICS · Observed versus expected information

For the Bernoulli, $J_n(\hat\theta)=k/\hat\theta^2+(n-k)/(1-\hat\theta)^2=n/(\hat\theta(1-\hat\theta))=I_n(\hat\theta)$: the two coincide at the MLE (true for every exponential family in its natural parametrization). For latent-variable models they differ; Efron and Hinkley (1978) favour $J_n(\hat\theta)$ for standard errors. EM does not return J_n, but Louis' (1982) identity recovers it from E-step quantities: $J_n(\hat\theta)=I_{\text{com}}-I_{\text{mis}}$,

where $I_{\text{com}} = \mathbb{E}\big[-\nabla^2 \log p(\mathbf{x}, \mathbf{h}; \theta) \mid \mathbf{x}\big]$ is the complete-data observed information averaged over the posterior of the hidden part $\mathbf{h}$, and $I_{\text{mis}} = \operatorname{Var}\big[\nabla \log p(\mathbf{x}, \mathbf{h}; \theta) \mid \mathbf{x}\big]$ is the posterior variance of the complete-data score, both at $\theta = \hat{\theta}$.

PART IV · STATISTICS

10.14 Asymptotics of the MLE and of CPT Entries

Theorem 10.8 (Consistency and asymptotic normality of the MLE)**.** *Under standard regularity conditions (identifiable model, θ_0 interior, three-times differentiable log-density with integrable bound, $I_1(\theta_0)$ positive definite), the MLE satisfies $\hat{\theta}_n \xrightarrow{p} \theta_0$ and $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N\big(0, I_1(\theta_0)^{-1}\big)$.*

Proof sketch. Consistency: $\frac{1}{n}\ell_n(\theta) \to \mathbb{E}_{\theta_0} \log p_\theta(X)$, which by Gibbs' inequality and identifiability is uniquely maximized at θ_0. Normality: expand the score, $0 = \nabla \ell_n(\hat{\theta}) = \nabla \ell_n(\theta_0) + \nabla^2 \ell_n(\tilde{\theta})(\hat{\theta} - \theta_0)$; the CLT gives $n^{-1/2}\nabla \ell_n(\theta_0) \xrightarrow{d} N(0, I_1)$, the LLN gives $-n^{-1}\nabla^2 \ell_n(\tilde{\theta}) \xrightarrow{p} I_1$, and Slutsky's lemma finishes (full proof: Casella and Berger, 2002, Section 10.1). □

For CPT entries the estimator is an explicit ratio of counts, and we can prove the result directly, including the role of the *random* number of records with parent configuration u.

Theorem 10.9 (Asymptotic normality of a CPT entry)**.** *Let the records be i.i.d. from a BN with $p_u = P(\mathrm{pa}(X) = u) > 0$, and $\hat{\theta}_{x|u} = N[x, u]/N[u]$. Then $\hat{\theta}_{x|u} \to \theta_{x|u}$ almost surely and*

$$\sqrt{N}\big(\hat{\theta}_{x|u} - \theta_{x|u}\big) \xrightarrow{d} N\left(0,\ \frac{\theta_{x|u}(1 - \theta_{x|u})}{p_u}\right).$$

Moreover, estimates belonging to different parent configurations $u \neq u'$ are asymptotically independent.

Proof. Write $\theta = \theta_{x|u}$ and $Z_j = \mathbb{1}\{U_j = u\}\big(\mathbb{1}\{X_j = x\} - \theta\big)$, where U_j is the parent configuration in record j. Then $\mathbb{E}Z_j = p_u\, \mathbb{E}[\mathbb{1}\{X = x\} - \theta \mid U = u] = 0$ and $\operatorname{Var} Z_j = \mathbb{E}Z_j^2 = p_u\, \theta(1 - \theta)$. Since $N[x, u] - \theta N[u] = \sum_j Z_j$,

$$\sqrt{N}(\hat{\theta} - \theta) = \frac{N^{-1/2} \sum_j Z_j}{N[u]/N}.$$

By the strong law, $N[u]/N \to p_u$ and $N^{-1}\sum_j Z_j \to 0$ a.s., giving consistency. By the CLT the numerator converges in distribution to $N(0, p_u\theta(1 - \theta))$; by Slutsky the ratio converges to $N(0, p_u\theta(1 - \theta)/p_u^2) = N(0, \theta(1 - \theta)/p_u)$. For $u \neq u'$, $Z_j Z_j' = 0$ because a record has only one parent configuration, so the numerators are uncorrelated; the multivariate CLT gives a jointly Gaussian limit with diagonal covariance, i.e. independence. □

The asymptotic variance $\theta(1 - \theta)/(Np_u)$ is the Bernoulli variance with the *effective sample size* Np_u: rare parent configurations are learned slowly, which is the statistical cost of large parent sets (Session 5's parameter blow-up seen through variances).

Numerical check (`ex_cpt_asymptotics()`, seed 8). Take binary $A \to B$ with $p_u = P(A{=}1) = 0.25$, $\theta = P(B{=}1 \mid A{=}1) = 0.6$ and $N = 400$ records. The asymptotic variance of $\hat{\theta} = N[B{=}1, A{=}1]/N[A{=}1]$ is $\theta(1-\theta)/(Np_u) = 0.24/100 = 0.002400$. Simulating 100 000 datasets (draw

$N[A{=}1] \sim \mathrm{Bin}(400, 0.25)$, then $N[B{=}1, A{=}1] \sim \mathrm{Bin}(N[A{=}1], 0.6)$) gives the variance 0.002413, close to the asymptotic value even though the effective sample size $N[A{=}1]$ (mean 100) is itself random.

PYTHON LAB · `s10_examples.py` · `ex_cpt_asymptotics()`

```
simulated Var (100000 replications, seed 8) = 0.002413, asymptotic =
0.24/100 = 0.002400.
```

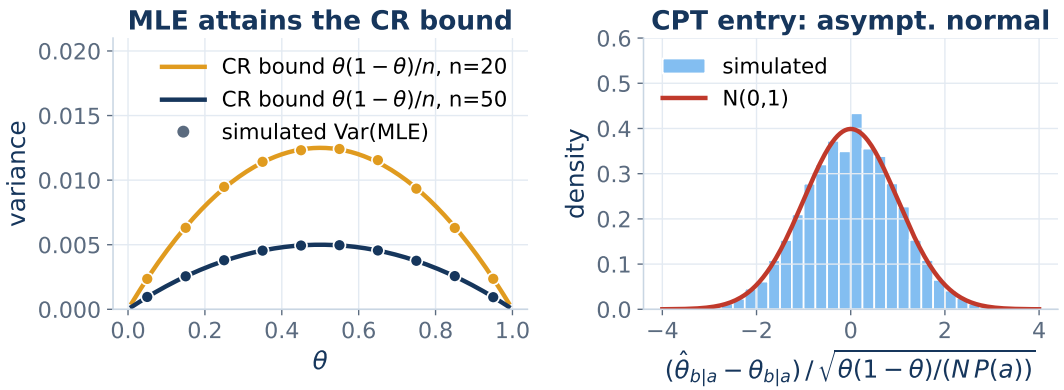


Figure 10.9. Left: variances of the Bernoulli MLE k/n, simulated with 40 000 samples at each of ten values of θ (dots), lie on the Cramér–Rao bound $\theta(1-\theta)/n$ (curves) for $n = 20$ and $n = 50$. Right: histogram of 60 000 simulated standardized CPT estimates of Theorem 10.9 ($N = 400$, $p_u = 0.25$, $\theta = 0.6$) against the $N(0,1)$ density (`fig_fisher()` in `s10_figures.py`).

PART IV · STATISTICS

10.15 Confidence Intervals for CPT Entries

Plugging $\hat\theta$ into Theorem 10.9 gives the *Wald* interval $\hat\theta \pm z\sqrt{\hat\theta(1-\hat\theta)/n}$ with $n = N[u]$. *Inverting* the score test instead, i.e. solving $|\hat\theta - \theta| \le z\sqrt{\theta(1-\theta)/n}$ for θ, gives the quadratic inequality $(1+\tfrac{z^2}{n})\theta^2 - (2\hat\theta + \tfrac{z^2}{n})\theta + \hat\theta^2 \le 0$, whose solution is the *Wilson* (1927) interval, always inside $[0,1]$:

$$\frac{\hat\theta + \frac{z^2}{2n}}{1+\frac{z^2}{n}} \;\pm\; \frac{z}{1+\frac{z^2}{n}}\sqrt{\frac{\hat\theta(1-\hat\theta)}{n} + \frac{z^2}{4n^2}}. \tag{10.4}$$

Its centre is itself a shrinkage estimator: for $z = 2$ it is $(k+2)/(n+4)$, the posterior mean under $\mathrm{Beta}(2,2)$ (the Agresti–Coull "add two successes and two failures" rule).

Worked Example 10.17 (Wald versus Wilson for a sparse CPT row)**.** An alarm network observes $k = 3$ alarms in $n = 20$ records with a given parent configuration, $\hat\theta = 0.15$; take $z = 1.96$.

1. *Wald.* $\sqrt{0.15 \cdot 0.85/20} = 0.07984$, half-width $1.96 \times 0.07984 = 0.1565$, interval $0.15 \pm 0.1565 = (-0.0065,\ 0.3065)$, which includes impossible negative values.
2. *Wilson*, by (10.4): $z^2/n = 3.8416/20 = 0.1921$, so the denominator is 1.1921; centre $= (0.15 + 0.0960)/1.1921 = 0.2460/1.1921 = 0.2064$; root $= \sqrt{0.006375 + 0.002401} = 0.09368$; half-width $= (1.96/1.1921) \times 0.09368 = 0.1540$. Interval $0.2064 \pm 0.1540 = (0.0524,\ 0.3604)$.

3. *Exact coverage* at the true value $\theta = 0.15$: $\sum_{k=0}^{20} \binom{20}{k} 0.15^k 0.85^{20-k} \mathbb{1}\{0.15 \in \mathrm{CI}(k)\}$. The Wald interval contains 0.15 only for $k = 2, \dots, 7$, the Wilson interval for $k = 0, \dots, 6$; summing the binomial probabilities gives 0.8185 (Wald) and 0.9781 (Wilson) for the nominal 0.95.

The Wald interval breaks down near 0 and 1, exactly where sparse CPT rows live (Brown, Cai and DasGupta, 2001).

PYTHON LAB · `s10_examples.py` · `ex_wald_wilson()`

```
Wald : (-0.0065, 0.3065); Wilson: z^2/n = 0.1921, denominator = 1.1921,
centre = 0.2064, half-width = 0.1540; Wilson : (0.0524, 0.3604); exact
coverage at theta=0.15, n=20: Wald 0.8185, Wilson 0.9781.
```

PART IV · STATISTICS

10.16 Bias, Shrinkage and Mean Squared Error

Proposition 10.8 (MSE of the Beta posterior mean). *For $k \sim \mathrm{Bin}(n, \theta)$ and $\tilde{\theta} = (k+\alpha)/(n+\alpha+\beta)$,*

$$\mathrm{MSE}(\tilde{\theta}) = \frac{n\theta(1-\theta) + \big(\alpha - (\alpha+\beta)\theta\big)^2}{(n+\alpha+\beta)^2}, \qquad \mathrm{MSE}(k/n) = \frac{\theta(1-\theta)}{n}.$$

Proof. $\mathbb{E}\tilde{\theta} = (n\theta+\alpha)/(n+\alpha+\beta)$, so the bias is $(\alpha-(\alpha+\beta)\theta)/(n+\alpha+\beta)$; $\mathrm{Var}\tilde{\theta} = \mathrm{Var}(k)/(n+\alpha+\beta)^2 = n\theta(1-\theta)/(n+\alpha+\beta)^2$. Use $\mathrm{MSE} = \mathrm{bias}^2 + \mathrm{Var}$. The MLE is the case $\alpha = \beta = 0$ (unbiased). □

Hence the MSE ratio tends to 1 as $n \to \infty$, while at the prior mean $\theta = \alpha/(\alpha+\beta)$ it equals $n^2/(n+\alpha+\beta)^2 < 1$ for every n.

Worked Example 10.18 (Shrinkage pays for small n). Apply Proposition 10.8 with Laplace's prior $\alpha = \beta = 1$, so $\tilde{\theta} = (k+1)/(n+2)$.

1. $n = 10$, $\theta = 0.3$: $\mathrm{MSE(MLE)} = 0.3 \cdot 0.7/10 = 0.021000$. For $(k+1)/12$: variance $10 \cdot 0.21/144 = 2.1/144$, bias $(1 - 2 \cdot 0.3)/12 = 0.4/12$, $\mathrm{bias}^2 = 0.16/144$, so $\mathrm{MSE} = (2.1 + 0.16)/144 = 0.015694$, a 25% reduction (ratio $0.015694/0.021 = 0.7474$).
2. Larger n at $\theta = 0.3$: $n = 100$ gives 0.002034 versus 0.002100 (ratio 0.9685); $n = 1000$ gives 0.000209 versus 0.000210 (ratio 0.9968).
3. Shrinkage can also hurt: at $\theta = 0.05$, $n = 10$, $\mathrm{MSE(MLE)} = 0.0475/10 = 0.004750$, while Laplace has variance $10 \cdot 0.0475/144$ and bias $(1-0.1)/12$, so $\mathrm{MSE} = (0.475+0.81)/144 = 0.008924$.

A Monte Carlo check with 400 000 draws of $k \sim \mathrm{Bin}(10, 0.3)$ (seed 4) gives 0.020975 and 0.015677.

PYTHON LAB · `s10_examples.py` · `ex_mse_mle_vs_bayes()`

```
exact MSE(MLE) = 0.021000, MSE(Bayes) = 0.015694, ratio = 0.7474; ratios
0.9685, 0.9968; theta=0.05,n=10: MSE(MLE)=0.004750, MSE(Bayes)=0.008924.
```

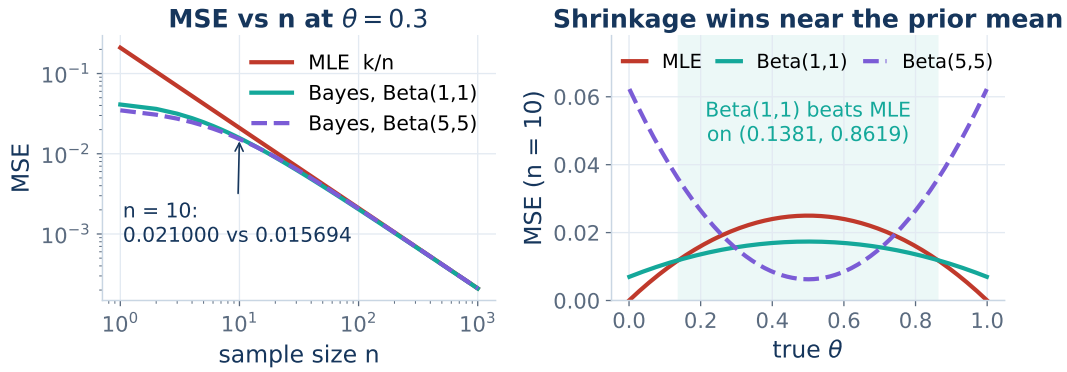


Figure 10.10. Exact MSEs from Proposition 10.8. Left: MSE against n at $\theta = 0.3$ for the MLE and the posterior means under Beta$(1,1)$ and Beta$(5,5)$ (log–log axes; the values at $n = 10$ are those of Worked Example 10.18). Right: MSE against θ at $n = 10$; the prior helps near its mean $1/2$ and hurts in the tails, the classic bias–variance trade-off. Beta$(1,1)$ beats the MLE exactly on $(0.1381,\ 0.8619)$, the interval where $\theta(1-\theta) > 10/84$: inserting $\alpha = \beta = a$ into Proposition 10.8, the inequality $\mathrm{MSE}(\tilde{\theta}) < \theta(1-\theta)/n$ simplifies to $\theta(1-\theta) > na/(4(n+a+na))$, which is $10/84$ for $n = 10$, $a = 1$.

PART IV · STATISTICS

10.17 EM Convergence and Identifiability

MATHEMATICAL STATISTICS · EM converges at the rate of the fraction of missing information

Near a fixed point θ^*, EM is a linear iteration $\theta^{(t+1)} - \theta^* \approx DM\,(\theta^{(t)} - \theta^*)$ with $DM = I_{\mathrm{com}}(\theta^*)^{-1} I_{\mathrm{mis}}(\theta^*) = \mathbf{I} - I_{\mathrm{com}}^{-1} I_{\mathrm{obs}}$ (Dempster, Laird and Rubin, 1977), where I_{com} is the expected complete-data information and $I_{\mathrm{mis}} = I_{\mathrm{com}} - I_{\mathrm{obs}}$. The linear rate is the largest eigenvalue of DM, the *largest fraction of missing information*.

Exact check. In Worked Example 10.9 keep only the complete and the B-missing records (drop the 20 A-missing ones). Among the 60 records with A=1, 40 are complete (30 with B=1) and 20 have B missing; the E-step gives each of the 20 the weight θ on B=1, so the EM update for $\theta = \theta_{B|1}$ is $\theta \mapsto (30 + 20\theta)/60$. Its fixed point is $30/40 = 0.75$, and $\frac{30+20\theta}{60} - 0.75 = \frac{20}{60}(\theta - 0.75)$: the error is multiplied by exactly $20/60 = 1/3$ per iteration (from $\theta^{(0)} = 0.5$ the errors are $0.083333,\ 0.027778,\ 0.009259, \ldots$). On the information side, one Bernoulli record carries $1/(\theta(1-\theta)) = 1/(0.75 \cdot 0.25)$, so with all 60 labels $I_{\mathrm{com}} = 60/0.1875 = 320$, with the 40 observed labels $I_{\mathrm{obs}} = 40/0.1875 = 213.33$, and $I_{\mathrm{mis}} = 320 - 213.33 = 106.67$: the fraction of missing information is $106.67/320 = 1/3$, the EM rate.

With the A-missing records added back, all three parameters are coupled. Differentiating the EM map numerically at the fixed point gives a 3×3 Jacobian with eigenvalues $0.1100,\ 0.3186,\ 0.4261$; the observed ratios of successive errors (error = largest absolute deviation of $\theta^{(t)}$ from the converged value) are $0.398, 0.406, 0.412, 0.415, 0.418, 0.428$ for $t = 1, \ldots, 6$ and $0.4282, 0.4262, 0.4261$ at $t = 10, 20, 25$, converging to the largest eigenvalue 0.4261.

PYTHON LAB · s10_examples.py · ex_em_rate(), ex_em_missing_cpt()

```
ratios:  [0.3333 ...]  (= 20/60);     I_com = 320.00, I_obs = 213.33, I_mis =
106.67, fraction missing = 0.3333;       eigenvalues of EM Jacobian:  0.1100,
0.3186, 0.4261; error ratio at t = 10, 20, 25:  [0.4282 0.4262 0.4261].
```

MATHEMATICAL STATISTICS · Label switching and identifiability of latent-variable models

A model is *identifiable* if $p_\theta = p_{\theta'}$ implies $\theta = \theta'$. A K-component mixture never is: permuting the component labels $(\pi_k, \mu_k, \sigma_k) \mapsto (\pi_{\rho(k)}, \mu_{\rho(k)}, \sigma_{\rho(k)})$ leaves p_θ unchanged, so the likelihood has $K!$ symmetric global maxima. For the customer data of Worked Example 10.11, $\ell = -882.7306$ at $\hat{\theta}$ and also at the label-swapped $\hat{\theta}$ (`ex_customer_segmentation()`). So compare EM restarts by ℓ, and never average parameters across runs without relabeling (e.g. impose $\mu_1 < \mu_2$). Beyond label switching, a latent-variable BN can be non-identifiable for dimensional reasons: a binary latent class C with two binary children X_1, X_2 has $1+2+2=5$ free parameters ($P(C{=}1)$ and one number per CPT row), but the observed table $P(X_1, X_2)$ has only $2^2 - 1 = 3$ degrees of freedom, so a continuum of parameter values fits equally well. With three binary children there are $1 + 2 \cdot 3 = 7$ parameters and $2^3 - 1 = 7$ observed degrees of freedom, so the dimension count no longer rules identifiability out; counting is a necessary condition only, not a proof of identifiability.

PART V

Learning in AI, Agents and Games

The same counts, priors and EM everywhere

PART V · AI CONNECTIONS

10.18 Connections to the Advanced AI Course

LINK TO COURSE SESSION 3 · Advanced Learning: cross-entropy is maximum likelihood

By Proposition 10.3, cross-entropy training is maximum likelihood. An autoregressive language model factorizes $p(w_{1:T}) = \prod_t p(w_t \mid w_{<t}; \theta)$: a Bayesian network on a complete DAG whose CPDs share one neural parametrization, and the pre-training loss is exactly $-\ell/(\text{number of tokens})$. Classical n-gram language models were literally CPT counting with Dirichlet-style smoothing (Worked Example 10.5).

LINK TO COURSE SESSION 4 · Generative Models: EM and the VAE

The VAE maximizes the same ELBO as Proposition 10.6, $\log p_\theta(x) \geq \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - \mathrm{KL}(q_\phi(z \mid x) \| p(z))$, but with a continuous latent z, a neural decoder and an *amortized* encoder q_ϕ in place of the exact E-step posterior (Kingma and Welling, 2014). EM closes the KL gap exactly; the VAE only shrinks it, so its bound is generally not tight.

AI CONNECTION · Reinforcement Learning

Model-based RL (course session 20) learns transition CPTs by counting; posterior sampling for RL draws a model from the Dirichlet posterior (Theorem 10.4), plans in it, and acts. A robot executes "move right" from state s twenty times (hypothetical log) and

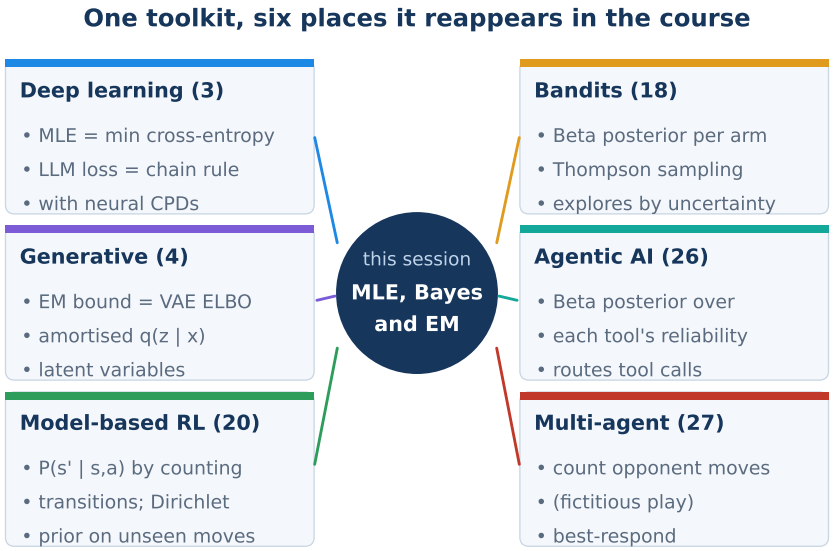


Figure 10.11. Where this session's estimators reappear (course session numbers in parentheses).

lands in the goal 14 times, slips and stays 5 times, and slides back once: counts $(14, 5, 1)$ over $s' \in (\text{goal}, \text{stay}, \text{back})$. The MLE transition row is $(14, 5, 1)/20 = (0.70, 0.25, 0.05)$; with a $\text{Dir}(1, 1, 1)$ prior the posterior mean is $(15, 6, 2)/23 = (0.6522, 0.2609, 0.0870)$. With values $V(s') = (1, 0, 0)$ (only the goal is worth anything), the one-step expected value $\sum_{s'} \hat{P}(s' \mid s, a)V(s')$ is 0.7000 under the MLE and 0.6522 under the posterior mean. By Theorem 10.9 the estimation error decays as $1/\sqrt{n_{s,a}}$: in a simulation that draws n transitions from the row $(0.7, 0.25, 0.05)$ 2000 times (seed 2), the mean of the largest absolute error $\max_{s'} |\hat{P}(s' \mid s, a) - P(s' \mid s, a)|$ is 0.0981, 0.0314, 0.0099 after $n = 20$, 200, 2000 visits, shrinking by about $\sqrt{10} = 3.16$ per tenfold increase.

PYTHON LAB · `s10_examples.py` · `ex_model_based_rl()`

```
MLE [0.7 0.25 0.05] -> E[V(s')] = 0.7000;  Dir(1) mean [0.6522 0.2609 0.087]
-> 0.6522; errors 0.0981, 0.0314, 0.0099.
```

LINK TO COURSE SESSION 18 · Exploration: Thompson sampling

Thompson sampling (Thompson, 1933) keeps a Beta posterior per arm (Theorem 10.2), samples one value per arm and pulls the arm with the largest sample, exploring as much as the posterior is uncertain. **Simulation** (`ex_thompson()`, `thompson_sim()`, seed 11): a Bernoulli bandit with three tools of success rates $(0.55, 0.65, 0.70)$, $T = 1000$ rounds, 300 independent runs, all agents starting from $\text{Beta}(1, 1)$ priors. The *cumulative regret* after T rounds is $\sum_{t=1}^{T}(p^* - p_{a_t})$ with $p^* = 0.70$: the expected number of successes lost by not always calling the best tool. Its mean over the 300 runs is 59.79 for greedy (always call the tool with the largest Laplace-smoothed mean $(S + 1)/(S + F + 2)$, S, F = successes, failures), 25.54 for ε-greedy ($\varepsilon = 0.1$: a uniformly random tool with probability 0.1, otherwise greedy) and 19.16 for Thompson sampling; see Fig. 10.12 and Russo et al. (2018).

PYTHON LAB · `s10_examples.py` · `ex_thompson()`

```
greedy :  mean cumulative regret at T=1000 = 59.79;       eps-greedy (0.1) :
25.54; Thompson :  19.16.
```

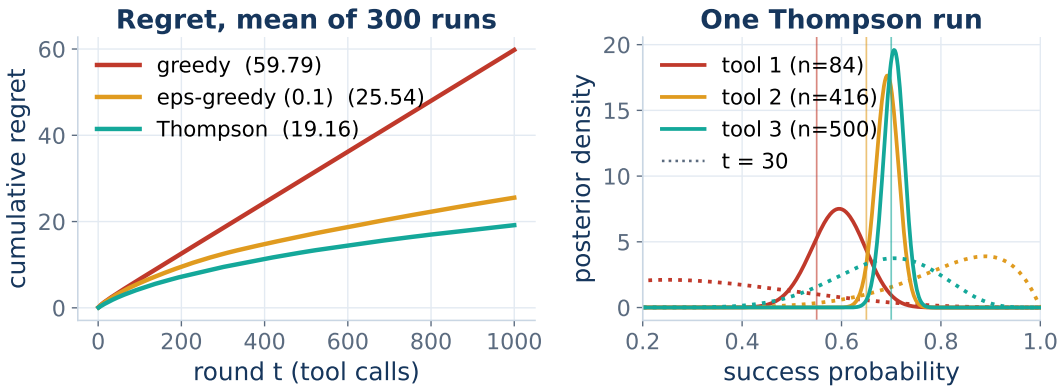


Figure 10.12. Left: mean cumulative regret of greedy, ε-greedy and Thompson sampling over the 300 simulated runs described above (final values in the legend). Right: Beta posteriors of one separate Thompson run (seed 5, `fig_thompson()` in `s10_figures.py`) after 30 (dotted) and 1000 (solid) rounds, with each tool's number of calls n in the legend; the best tool (tool 3, rate 0.70) is pulled most and its posterior is the sharpest; vertical lines mark the true rates.

AI CONNECTION · Agentic AI

An agent that learns which tool to trust. An LLM assistant can call two search tools. With uniform Beta$(1,1)$ priors, tool A succeeded 18 times in 20 calls and tool B 4 times in 4: posteriors Beta$(19,3)$ (mean 0.8636, 95% interval $(0.6962, 0.9695)$) and Beta$(5,1)$ (mean 0.8333, interval $(0.4782, 0.9949)$). (The intervals are 95% equal-tailed credible intervals: the 2.5% and 97.5% Beta quantiles.) Thompson sampling draws θ_A, θ_B independently from the two posteriors and calls B iff $\theta_B > \theta_A$. Beta$(5,1)$ has cdf $P(\theta_B \le t) = t^5$, so conditioning on θ_A gives $P(\theta_B > \theta_A) = \mathbb{E}[1 - \theta_A^5]$; with the Beta$(a,b)$ moments $\mathbb{E}[\theta^m] = \prod_{i=0}^{m-1} \frac{a+i}{a+b+i}$,

$$P(\theta_B > \theta_A) = 1 - \frac{19 \cdot 20 \cdot 21 \cdot 22 \cdot 23}{22 \cdot 23 \cdot 24 \cdot 25 \cdot 26} = 1 - \frac{7980}{15600} = 1 - 0.5115 = 0.4885.$$

B has the lower mean but is still tried almost half of the time because it is so uncertain (a Monte Carlo check with 10^6 draws, seed 9, gives 0.4881); a greedy agent would always call A and never test B again.

PYTHON LAB · `s10_examples.py` · `ex_tool_reliability()`

```
E[theta_A^5] = 7980/15600 = 0.5115;        tool A: Beta(19,3) mean=0.8636 95%
CI=[0.6962 0.9695];     tool B: Beta(5,1) mean=0.8333 95% CI=[0.4782 0.9949];
P(theta_B > theta_A) exact = 0.4885, Monte Carlo = 0.4881.
```

LINK TO COURSE SESSION 26 · Agentic AI: memory as sufficient statistics

An agent's memory about its tools need not be a transcript: the counts are sufficient (Proposition 10.2) and updates are $O(1)$ (Proposition 10.4). Missing outcome labels in agent logs are a missing-data problem (Table 10.2).

AI CONNECTION · Multi-Agent AI

Fictitious play learns an opponent model by counting. In rock–paper–scissors (payoff $+1$ for a win, -1 for a loss, 0 for a tie), an agent has seen its opponent play R, P, S $12, 5, 3$ times. Its Dirichlet$(1,1,1)$ predictive for the opponent's next move is $(13,6,4)/23 = (0.5652, 0.2609, 0.1739)$. Expected payoffs: R beats S and loses to P, so

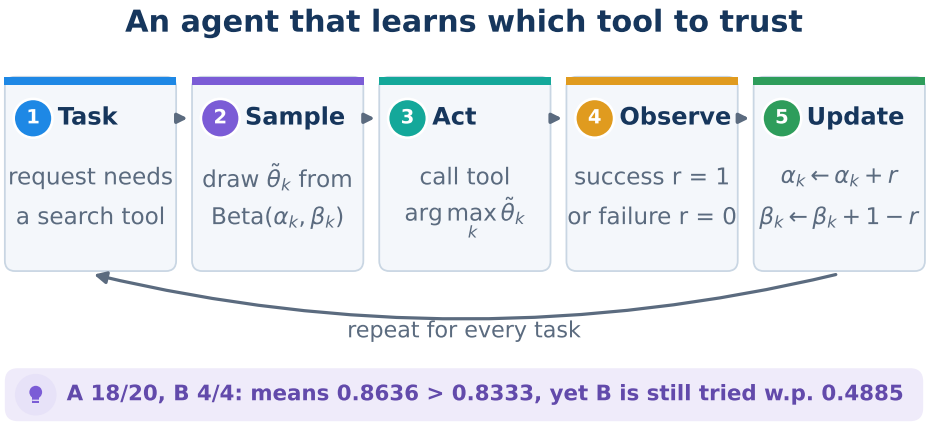


Figure 10.13. The tool-reliability loop of an agent: sample from each tool's Beta posterior, act greedily on the samples, observe success, add the outcome to the counts (Proposition 10.4).

$\mathbb{E}[\mathrm{R}] = 0.1739 - 0.2609 = -0.0870$; similarly $\mathbb{E}[\mathrm{P}] = 0.5652 - 0.1739 = 0.3913$ and $\mathbb{E}[\mathrm{S}] = 0.2609 - 0.5652 = -0.3043$. The best response is Paper. When two agents both play this way, each counting the other's moves with a $\mathrm{Dir}(1,1,1)$ prior and best-responding to its predictive (fictitious play, Brown 1951; `fictitious_play_rps()`), the empirical frequencies of their moves converge to the Nash equilibrium $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ in this zero-sum game (Robinson, 1951); after 5000 rounds the second player's frequencies of R, P, S are $(0.328, 0.333, 0.339)$.

PYTHON LAB · `s10_examples.py` · `ex_fictitious_play()`

```
Dir(1) predictive = [0.5652 0.2609 0.1739];  expected payoff R,P,S = [-0.087
0.3913 -0.3043]; best response = P; FP frequencies = [0.328 0.333 0.339].
```

LINK TO COURSE SESSION 27 · Multi-Agent AI: learning about other agents

Opponent modelling and Bayesian games keep a posterior over other agents' strategies (Dirichlet–multinomial) or hidden *types* (a latent-variable model fitted by EM). Agents can pool counts; by Proposition 10.4 the order of reports is irrelevant.

Table 10.3. This session in the Advanced AI course.

Concept	Where it reappears (course session / notes Session)	Role there
MLE / cross-entropy	course 3, 31	loss of every classifier and LLM
Bayesian updating	course 18, 21	Thompson sampling; POMDP beliefs
Dirichlet counts	course 20, 27	transition and opponent models
EM / ELBO	course 4; Session 11	VAEs; variational inference
Missing data	course 26; Sessions 12–15	agent logs; ignorability
Fisher information	course 19	natural policy gradient

Treat the potential outcome $Y(1)$ (the outcome a unit would have under treatment, Session 15) as missing for untreated units: "MAR given covariates" is the ignorability assumption of Session 15, and complete-case bias is confounding bias.

PART V · AI CONNECTIONS

10.19 Summary

Table 10.4. Cheat sheet of key formulas.

Object	Formula
Decomposition	$\ell(\theta) = \sum_i \sum_u \sum_x N[x,u] \log \theta_{x\|u}$
MLE	$\hat{\theta}_{x\|u} = N[x,u]/N[u]$
Dirichlet posterior	$\theta_{\cdot\|u} \mid \mathcal{D} \sim \mathrm{Dir}(\boldsymbol{\alpha}_u + \mathbf{N}_u)$; predictive $(\alpha_{x\|u} + N[x,u])/(\alpha_u + N[u])$
ELBO	$\log p(x;\theta) = \mathcal{L}(q,\theta) + \mathrm{KL}(q\|p(\cdot \mid x;\theta))$
GMM M-step	$\pi_k = N_k/N$, $\mu_k = \sum_j r_{jk} x_j / N_k$, $\sigma_k^2 = \sum_j r_{jk}(x_j - \mu_k)^2 / N_k$
Expected counts	$\bar{M}[x,u] = \sum_j P(x,u \mid \mathbf{o}^{(j)}; \theta^{(t)})$
Fisher / CR	$I_1 = 1/(\theta(1-\theta))$; $\mathrm{Var}(T) \geq g'(\theta)^2 / I_n(\theta)$
CPT asymptotics	$\sqrt{N}(\hat{\theta}_{x\|u} - \theta_{x\|u}) \xrightarrow{d} N(0, \theta(1-\theta)/p_u)$
Shrinkage MSE	$[n\theta(1-\theta) + (\alpha - (\alpha+\beta)\theta)^2]/(n+\alpha+\beta)^2$
EM rate	largest eigenvalue of $I_{\mathrm{com}}^{-1} I_{\mathrm{mis}}$

Table 10.5. Common pitfalls.

Pitfall	Why it hurts	Remedy
Zero counts	probability 0, $\ell = -\infty$ on new data	Dirichlet prior / smoothing
Wald interval for sparse rows	coverage far below nominal, negative bounds	Wilson or Bayesian interval
Complete-case analysis	biased unless MCAR	likelihood/EM under MAR; model R if MNAR
Single EM run	local optima, symmetric fixed points	several random restarts, keep best ℓ
Collapsing GMM component	unbounded likelihood	variance floor, MAP-EM

THINK ABOUT IT An LLM is trained by maximum likelihood on internet text. Which of the missing-data mechanisms best describes the text that is *not* on the internet, and what does that imply for the model?

THINK ABOUT IT EM and a VAE both maximize an ELBO. What exactly is lost when the E-step is amortized by an encoder network?

THINK ABOUT IT A tool-using agent only logs a success label when the user clicks "thumbs up". Is the reliability estimate of Fig. 10.13 still valid?

Further reading. Koller and Friedman (2009), *Probabilistic Graphical Models*, Ch. 17 (parameter estimation) and Ch. 19 (partially observed data); Murphy (2022), Ch. 4; Bishop (2006), *Pattern Recognition and Machine Learning*, Ch. 9 (mixtures and EM); Casella and Berger (2002), Ch. 7 and

10 (MLE, Cramér–Rao, asymptotics); Little and Rubin (2002), *Statistical Analysis with Missing Data*; Dempster, Laird and Rubin (1977), *JRSS-B* 39; Sutton and Barto (2018), Ch. 2, 8.

APPENDIX

10.20 Appendix: Proofs

Proof of EM Monotonicity (Theorem 10.6)

Let $q_j^{(t)}(c) = P(c \mid x^{(j)}; \theta^{(t)})$ and let Q and H be as in (10.2), so $H(q^{(t)}) = -\sum_j \sum_c q_j^{(t)}(c) \log q_j^{(t)}(c)$ does not depend on θ. Applying Proposition 10.6 to every record (with $q = q_j^{(t)}$ and an arbitrary θ) and summing over j,

$$\ell(\theta; \mathcal{D}) = Q(\theta \mid \theta^{(t)}) + H(q^{(t)}) + \sum_j \mathrm{KL}\big(q_j^{(t)} \,\|\, P(\cdot \mid x^{(j)}; \theta)\big) \qquad \text{for every } \theta. \qquad (*)$$

1. *The bound touches at $\theta^{(t)}$.* For $\theta = \theta^{(t)}$ each KL term compares $q_j^{(t)}$ with itself and is 0, so $\ell(\theta^{(t)}; \mathcal{D}) = Q(\theta^{(t)} \mid \theta^{(t)}) + H(q^{(t)})$.
2. *The bound lies below ℓ.* $\mathrm{KL} \geq 0$ (Gibbs' inequality: $-\mathrm{KL}(q\|p) = \sum q \log(p/q) \leq \log \sum q\,(p/q) = \log 1 = 0$ by Jensen's inequality for the concave log), so by $(*)$, $\ell(\theta; \mathcal{D}) \geq Q(\theta \mid \theta^{(t)}) + H(q^{(t)})$ for every θ.
3. *The M-step climbs the bound.* $\theta^{(t+1)} = \arg\max_\theta Q(\theta \mid \theta^{(t)})$, so $Q(\theta^{(t+1)} \mid \theta^{(t)}) \geq Q(\theta^{(t)} \mid \theta^{(t)})$.

Chaining 2, 3 and 1:

$$\ell(\theta^{(t+1)}; \mathcal{D}) \;\geq\; Q(\theta^{(t+1)} \mid \theta^{(t)}) + H(q^{(t)}) \;\geq\; Q(\theta^{(t)} \mid \theta^{(t)}) + H(q^{(t)}) \;=\; \ell(\theta^{(t)}; \mathcal{D}).$$

The same argument works for any $\theta^{(t+1)}$ that merely increases Q (generalized EM).

Expected counts are the E-step of a BN (derivation of (10.3))

The complete-data log-likelihood of record j is $\sum_i \sum_{x,u} \mathbb{1}\{x_i^{(j)}{=}x, \mathrm{pa}_i^{(j)}{=}u\} \log \theta_{x|u}$. Its expectation under $P(\mathbf{h}^{(j)} \mid \mathbf{o}^{(j)}; \theta^{(t)})$ replaces each indicator by its posterior probability, so $Q(\theta \mid \theta^{(t)}) = \sum_{i,x,u} \bar{M}_t[x,u] \log \theta_{x|u}$, the form (10.1) with real-valued counts; Theorem 10.1 (whose proof never used integrality) gives the M-step.

CONNECTIONS AND TAKEAWAYS

We can now learn CPT parameters exactly when data is fully observed (a direct, decomposable payoff of the DAG factorization), and, with latent variables, up to a local maximum of the likelihood by alternating inference (Session 7) with the same decomposable MLE (EM). Two loose threads remain before this toolkit is genuinely practical. First, we have always assumed the graph structure G itself is known — but where does the graph come from when domain knowledge is incomplete? Second, the E-step's inference call assumed variable elimination is

tractable, but Session 7 showed VE's cost explodes with treewidth — what happens when it is not tractable? **Session 11** addresses both: a brief look at learning structure from data (where the Dirichlet marginal likelihood of today's Part II becomes the Bayesian score), and a full treatment of sampling-based approximate inference (including Markov chain Monte Carlo) for exactly the cases where exact inference is computationally out of reach.